Sasakian Hodge Numbers of Calabi–Yau Links

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Abstract

The link of an isolated weighted homogeneous hypersurface singularity whose degree equals the sum of its weights is a Sasakian manifold that admits a contact Calabi–Yau structure, and its Sasakian Hodge numbers are dimensions of graded pieces of the Milnor algebra of the defining polynomial. In earlier work these numbers were computed with Gröbner bases for the 7-dimensional links of 7549 of the 7555 weighted Calabi–Yau threefold links, and two conjectures were raised from the data. The first asserts that polynomials with the same weights have Milnor algebras whose relevant graded pieces have equal dimensions; the second asserts that the Sasakian Hodge number $h^{2,1}_S$ of the link is bounded above by the Hodge number $h^{2,1}$ of the base Calabi–Yau threefold. Both conjectures are proved here. The first holds for every polynomial with an isolated singularity, because its partial derivatives form a regular sequence; an independent derivation of the Sasakian Hodge numbers by sheaf cohomology on the leaf-space orbifold is also given. For the second, the stringy Hodge numbers of the Calabi–Yau orbifold are shown to bound the middle Sasakian Hodge numbers in every dimension $n \geq 2$, and for $n \geq 3$ the gap in $h^{n-1,1}$ is shown to be a sum of geometric genera of loci of orbifold points; for threefolds the bound is attained exactly when all curves of orbifold points are rational. The resulting closed formulae give the Sasakian Hodge numbers of the six 7-dimensional links left open in the earlier work and complete datasets for all 95 five-dimensional K3 links and all 1 100 055 nine-dimensional Calabi–Yau fourfold links; for the K3 links they recover the previously known second Betti numbers.

Code and data: github.com/edhirst/Link_SasakianHodgeNumbers.

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1 Introduction

Weighted hypersurface singularities provide a common setting for Sasakian geometry, Hodge theory and the topology of links. The problem considered here is to determine which Hodge-theoretic invariants of a Calabi–Yau link are fixed by the weights of its defining equation, and how they compare with those of the corresponding Calabi–Yau hypersurface. A weighted homogeneous polynomial f in N complex variables with an isolated singularity at the origin determines two closely related geometric objects. The first is the hypersurface $X_f = \{f = 0\}$ in a weighted projective space $\mathbb{P}^{N-1}(\mathbf{w})$, a projective variety with cyclic quotient singularities, which is the coarse space of a compact complex orbifold $\mathcal{X}_f$ of dimension $n = N-2$. The second is the *link* $K_f = f^{-1}(0) \cap S^{2N-1}$, a smooth compact manifold of dimension $2n+1$, which is a Seifert circle orbibundle over $\mathcal{X}_f$. The link inherits a quasi-regular Sasakian structure whose Reeb flow is the weighted circle action. When the degree of f equals the sum of the weights, the orbifold $\mathcal{X}_f$ has trivial canonical bundle, and after a deformation of its transverse Kähler metric the link carries a *contact Calabi–Yau* structure [TV2008, HV2015, CRS2020]. In real dimension 7 such structures induce G_2-structures [HV2015, CRS2020], which is one of the reasons for interest in them [LS2023].

The *Sasakian Hodge numbers* $h_S^{p,q}$ of the link are the dimensions of the spaces of harmonic forms of type (p,q) in degrees up to the middle dimension. Apart from $h_S^{0,0} = 1$ they vanish below the middle degree, because the link is highly connected (Remark 3.14). The remaining ones give a Hodge decomposition of the middle cohomology of K_f, and in particular determine its Betti number, but not finer invariants such as torsion or the diffeomorphism type. By a theorem going back to Steenbrink and Itoh [S1977a, I2004], they are the dimensions of certain graded pieces of the Milnor algebra $\mathbb{M}_f = \mathbb{C}[z]/(\partial_1 f, \dots, \partial_N f)$. In [AHHH+2024a] this description was turned into an algorithm based on Gröbner bases, to compute the Sasakian Hodge numbers and the Crowley–Nordström invariant [CN2015] of the 7-dimensional links for 7549 of the 7555 weighted Calabi–Yau threefolds [CLS1990, KS1994, KS1992a]; then studied with machine learning. Two conjectures were raised from the data. Conjecture 7 of [AHHH+2024a], referred to here as the *Weight Determinacy Conjecture*, states that two polynomials with the same weights have Milnor algebras whose graded pieces in the relevant degrees have equal dimensions, so that the invariants of the link depend only on the weights. Conjecture 8 of [AHHH+2024a], referred to here as the *Calabi–Yau Bound Conjecture*, states that the Sasakian Hodge number $h_S^{2,1}$ of a 7-dimensional link never exceeds the Hodge number $h_{CY}^{2,1}$ of the corresponding Calabi–Yau threefold; equality was observed in 4198 of the 7549 cases.

Definition 1.1 (Admissible weighted hypersurfaces)**.** Let $\mathbf{w} = (w_1, \dots, w_N)$ be positive integer weights, and give $S = \mathbb{C}[z_1, \dots, z_N]$ the grading $\deg z_i = w_i$. A polynomial $f \in S_d$ is *admissible* if it has an isolated critical point at the origin. For such an f, retain the notation K_f, $\mathcal{X}_f$ and X_f for its link, quotient orbifold and coarse hypersurface, put $n = N-2$, and write

$$Y = f^{-1}(0) \setminus 0, \qquad \mathbb{M}_f = S/(\partial_1 f, \dots, \partial_N f).$$

The data are *Calabi–Yau* when $d = \sum_i w_i$. The monomial counts and the formal coefficients a_k are defined by

$$p_{\mathbf{w}}(k) = \dim S_k, \qquad P_{\mathbf{w}}(t) = \prod_{i=1}^{N} \frac{1 - t^{d-w_i}}{1 - t^{w_i}} = \sum_{k \geq 0} a_k t^k,$$

with $p_{\mathbf{w}}(k) = 0$ for $k < 0$.

We prove both conjectures and relate them to the classical descriptions of the invariants involved. The algebraic input is the Hilbert-series calculation for a weighted complete intersection, underlying the Milnor–Orlik formula [MO1970, S1978]; a recent account of the Koszul-complex proof is given by Soler [S2026]. The Weight Determinacy Conjecture is a consequence of this classical calculation. When f has an isolated singularity its partial derivatives form a regular sequence. With the notation of Definition 1.1, this gives

$$\sum_{\ell \geq 0} \dim(\mathbb{M}_f)_\ell \, t^\ell = \prod_{i=1}^{N} \frac{1 - t^{\,d-w_i}}{1 - t^{\,w_i}} \,. \tag{1.1}$$

This gives the statement in every degree, not only in those relevant to the Hodge numbers, once the isolated-singularity hypothesis implicit in [AHHH+2024a] is made explicit; Example 3.3 shows that the hypothesis

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cannot be dropped. Two polynomials are weakly R-equivalent when their Milnor algebras have graded pieces of equal dimensions in the degrees used for the Sasakian Hodge numbers (Section 2.3).

Theorem A (Weight determinacy)**.** *Let f and g be admissible polynomials in N variables, in the sense of Definition 1.1, of the same degree d and with weight vectors agreeing up to permutation. Then their graded Milnor algebras satisfy*

$$\dim(\mathbb{M}_f)_\ell = \dim(\mathbb{M}_g)_\ell \qquad (\ell \in \mathbb{Z}).$$

In particular f and g are weakly R-equivalent in the sense of Section 2.3, and they have the same Sasakian Hodge numbers and the same Milnor number. If moreover $N = 5$ and $d = \sum_i w_i$, they have the same value of (2.12), and hence the same Crowley–Nordström invariant.

Theorem A also answers the question posed by Itoh [I2004, p. 123, Problem following Cor. 1.3]: whether the Sasakian Hodge numbers depend only on the weights and degree, at least for five-dimensional links. The answer holds in every dimension $2n + 1 \geq 5$ considered here, without the Calabi–Yau condition. At the algebraic level, (1.1) refines the Milnor–Orlik product formula by retaining the grading; evaluating at $t = 1$ recovers the Milnor number [MO1970, S2026].

The next theorem computes the basic Hodge numbers of the link by sheaf cohomology on the leaf-space orbifold and reproduces the coefficients of (1.1).

Theorem B (Sasakian Hodge numbers by sheaf cohomology)**.** *Let f be admissible as in Definition 1.1, with $n \geq 2$. Use the monomial counts $p_{\mathbf{w}}$ and coefficients a_k from that definition. Then $h_S^{0,0}(K_f) = 1$, $h_S^{p,q}(K_f) = 0$ for $0 < p + q < n$, and for $p + q = n$*

$$h_S^{p,q}(K_f) = \sum_{\substack{I \subseteq \{1,\dots,N\} \\ |I| \leq p}} (-1)^{|I|}\, p_{\mathbf{w}}\Big((p+1)d - \sum_i w_i - \sum_{i \in I}(d - w_i)\Big) = a_{(p+1)d - \sum_i w_i}. \tag{1.2}$$

Combining Theorem B with the classical Hilbert-series identity (1.1) recovers Itoh's formula for the Sasakian Hodge numbers in terms of the Milnor algebra [I2004, Thm. 1.2]. Itoh's proof uses the residue-theoretic description of primitive cohomology due to Griffiths and Steenbrink [G1969a, G1969b, S1977a]. The contribution here is an alternative sheaf-cohomological derivation under the same assumptions, using neither residues nor the Jacobian-ring description of the Hodge structure.

In the Calabi–Yau case the outcome is a closed formula in terms of the numbers $p_{\mathbf{w}}(k)$ of monomials of weighted degree k; for 7-dimensional links, for instance,

$$h_S^{2,1} = p_{\mathbf{w}}(d) - \sum_{i=1}^{5} p_{\mathbf{w}}(w_i)\,. \tag{1.3}$$

The second conjecture concerns the relation between ordinary orbifold cohomology and stringy Hodge theory. The relevant framework is provided by Chen–Ruan cohomology [CR2004], Batyrev's stringy invariants [B1998] and their comparison by Yasuda [Y2004]. The proof compares the Hodge numbers in Table 1.1; in particular, the ordinary Hodge numbers of the quotient orbifold $\mathcal{X}_f$ must be distinguished from its Chen–Ruan Hodge numbers.

Table 1.1: Hodge-number conventions for the link and its quotient.

Notation	Meaning
$h_B^{p,q}(K_f) = h^{p,q}(\mathcal{X}_f)$	Basic Hodge numbers of the link, equal to the ordinary Hodge numbers of the quotient orbifold.
$h_S^{p,q}(K_f)$	Sasakian Hodge numbers: the primitive part of basic cohomology in degrees up to the middle.
$h_{\mathrm{orb}}^{p,q}(\mathcal{X}_f)$	Chen–Ruan Hodge numbers, including contributions from twisted sectors.
$h_{CY}^{p,q} = h_{st}^{p,q}(X_f)$	Stringy Hodge numbers of the coarse hypersurface, equal to $h_{\mathrm{orb}}^{p,q}(\mathcal{X}_f)$.

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The Kreuzer–Skarke lists record the stringy Hodge numbers. In the middle degree, the comparison is

$$h^{p,q}_{CY} = h^{p,q}_{B} + \big(\text{twisted contributions}\big), \qquad h^{p,q}_{B} = h^{p,q}_{S} + \delta_{pq} \qquad (p+q=n). \tag{1.4}$$

The non-negativity of the twisted contributions gives the bound in every dimension $n \geq 2$. The following theorem identifies the gap in $h^{n-1,1}$ for $n \geq 3$. Here μ_r is the group of r-th roots of unity, and for an index set F the notation $p_{\mathbf{w}_F}$ counts monomials in the variables indexed by F, with their inherited weights.

Theorem C (The Calabi–Yau bound and its defect)**.** *Let f be an admissible Calabi–Yau polynomial as in Definition 1.1, and put $X = X_f$ and $h^{p,q}_{CY} = h^{p,q}_{st}(X)$. For $F \subseteq \{1,\dots,N\}$ write $\mathbf{w}_F = (w_i)_{i\in F}$ and $g_F = \gcd(w_i : i \in F)$, and set $p_{\mathbf{w}_F}(k) = 0$ for $k < 0$. Let $p+q = n$.*

(i) For $n \geq 2$, $h^{p,q}_{S}(K_f) \leq h^{p,q}_{CY} - \delta_{pq}$.

(ii) For $n \geq 3$,

$$h^{n-1,1}_{CY} - h^{n-1,1}_{S}(K_f) = \sum_{\substack{|F|=n \\ g_F>1}} (g_F - 1)\, p_{\mathbf{w}_F}\Big(d - \sum_{i\in F} w_i\Big). \tag{1.5}$$

The summand indexed by F is $g_F - 1$ times the geometric genus $p_g = h^{n-2,0}$ of the closed suborbifold $\mathcal{C}_F = [(Y \cap \mathbb{C}^F)/\mathbb{C}^*]$ of points whose stabiliser contains μ_{g_F}, whose coarse space is $X \cap \mathbb{P}(\mathbf{w}_F)$.

The loci $\mathcal{C}_F$ in Theorem C have dimension $n-2$ and codimension two in X_f. For threefolds they are curves, and the bound is attained exactly when all of them are rational.

Theorem D (The threefold bound)**.** *Under the assumptions and notation of Theorem C, let $n = 3$, so that X_f is a threefold and K_f is seven-dimensional. For each index set F with $|F| = 3$ and $g_F > 1$, the coarse space $C_F = X_f \cap \mathbb{P}(\mathbf{w}_F)$ of $\mathcal{C}_F$ is a smooth connected projective curve. Its generic points have stabiliser μ_{g_F} and are transversal singularities of type A_{g_F-1} of X_f, while finitely many special points may have larger stabilisers. With the geometric genus $p_g(C_F) = p_{\mathbf{w}_F}(d - \sum_{i\in F} w_i)$ of the whole curve,*

$$h^{2,1}_{CY} = h^{2,1}_{S}(K_f) + \sum_{|F|=3,\ g_F>1} (g_F - 1)\, p_g(C_F) \ \geq\ h^{2,1}_{S}(K_f).$$

Equality holds if and only if every such curve is rational, in particular when there is none.

Theorems C and D express the classical distinction between polynomial and non-polynomial deformations [GH1987, COK1995] in terms of the Sasakian Hodge numbers of the link. They give a geometric interpretation and an exact error formula for the $h^{n-1,1}$ approximation of [HS2024, §5], recovering its lower-bound property in dimensions three and four and giving the corresponding statement for every $n \geq 3$. Remark 4.5 makes these comparisons explicit. Equality holds for 4202 of the 7555 weight systems, and for 4198 of the 7549 considered in [AHHH+2024a]. The same decomposition explains why the Sasakian Hodge numbers are not mirror symmetric, as the link sees only the untwisted sector, while mirror symmetry exchanges untwisted with twisted contributions (Remark 4.7). For fourfolds the same decomposition gives a Libgober–Wood identity, which explains the near-linear relation between the two Sasakian Hodge numbers seen in the data. For the punctured cone Y of Definition 1.1 and a finite-order element $t \in \mathbb{C}^*$, let $Y^t = \{z \in Y : t \cdot z = z\}$ and $\mathcal{X}_t = [Y^t/\mathbb{C}^*]$. The non-empty sectors with $t \neq 1$ are the twisted sectors. Their age is the sum of the fractional arguments of the eigenvalues of t on the tangent space of $\mathcal{X}_f$, with arguments taken in $[0,1)$ (Lemma 4.1).

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Theorem E (Libgober–Wood identity for nine-dimensional links)**.** *Under the assumptions of Theorem C, let $n = 4$ and use the sectors $\mathcal{X}_t$ defined above. The middle Sasakian Hodge numbers of the nine-dimensional link K_f satisfy*

$$h_S^{2,2} = 4h_S^{3,1} + 47 + R, \qquad R = \sum_{t \neq 1} \rho(t), \tag{1.6}$$

where t runs over the non-trivial sectors $\mathcal{X}_t$ defined above. If $\mathcal{X}_t$ has age a and dimension m, then $m \leq 2$, and its contribution is given by

Dimension m	*Age a*	*Contribution $\rho(t)$*
2	—	$6\chi(\mathcal{O}_{\mathcal{X}_t}) - e(\mathcal{X}_t)$
1	1	$e(\mathcal{X}_t) + 1$
1	2	-1
0	1	$4c$
0	2	$-c$
0	3	0

where $\chi(\mathcal{O}_{\mathcal{X}_t})$ is the holomorphic Euler characteristic, e is the Euler characteristic of the coarse space and c the number of points of a sector of dimension zero. The sectors of dimension one occur in pairs $\{t, t^{-1}\}$ of ages 1 and 2, and each pair contributes $e(\mathcal{X}_t)$. In particular $h_S^{2,2} = 4h_S^{3,1} + 47$ when X_f has no orbifold points.

Theorem E is a consequence of Batyrev's stringy Libgober–Wood identity [B2000, Cor. 3.10, Cor. 3.11], after separating the untwisted and twisted contributions. It gives the resulting relation on the link together with the correction contributed by each sector.

The closed formulae express each Sasakian Hodge number considered here as an alternating sum of at most 22 denumerants $p_{\mathbf{w}}(k)$, which are computed exactly in milliseconds. In the first place they complete the 7-dimensional dataset of [AHHH+2024a], via the invariants of the six weight systems whose Gröbner bases could not be computed as given in Section 3.4, where they are also certified exactly for explicit polynomials. They also give two further datasets. The first consists of the 5-dimensional links of the 95 weighted K3 surfaces. Through $b_2(K_f) = 2 + h_S^{1,1}(K_f)$, it recovers the Betti numbers computed by Boyer, Galicki and Matzeu [BGM2006, Ex. 41, Cor. 42] and gives a uniform Hodge-theoretic calculation with independent Gröbner-basis checks; the full list is given in Table A.2. The second consists of the 9-dimensional links of all 1 100 055 weighted Calabi–Yau fourfolds [LSW1999]. The Gröbner-basis approach of [AHHH+2024a] enumerates a monomial basis of the whole Milnor algebra, whose dimension reaches 3×10^{12} in this dataset, and it originally did not complete under months of CPU compute for these links; with the closed formulae all of them are processed in about five minutes on a laptop. The results are checked by four verification methods, including exact certificates for explicit polynomials and a comparison with the Kreuzer–Skarke data, and they are summarised in Tables 6.1 and 6.2.

This paper is organised as follows. Section 2 recalls the background, fixes the conventions for the orbifold and its coarse space, and restates the two conjectures. Section 3 proves Theorems A and B and derives the closed formulae. Section 4 proves Theorems C, D, and E. Sections 5 and 6 discuss the data for 5- and 9-dimensional links, and Section 7 concludes. Appendix A records the computational certificates, the computing environment and the complete K3 table. The data, code and certificates are available in the corresponding GitHub repository.

2 Background

We fix the weighted grading and the conventions for the link K_f, the quotient orbifold $\mathcal{X}_f$, and its coarse space X_f. The distinction between basic, Sasakian, and stringy Hodge numbers is essential for the comparison outlined in Section 1. After recalling the Milnor-algebra description of the Sasakian Hodge numbers, we state the two conjectures in this notation.

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2.1 Weighted homogeneous polynomials and Milnor algebras

Throughout, $N = n + 2 \geq 4$, and $\mathbf{w} = (w_1, \dots, w_N)$ is a vector of positive integer weights with $\gcd(\mathbf{w}) = 1$; when weight systems are listed or counted, their weights are ordered nondecreasingly. The polynomial ring $S = \mathbb{C}[z_1, \dots, z_N]$ is graded by $\deg z_i = w_i$, and S_k denotes its degree-k part. The dimension of S_k is the *denumerant*

$$p_{\mathbf{w}}(k) = \#\Big\{ m \in \mathbb{Z}_{\geq 0}^N : \sum_{i=1}^N m_i w_i = k \Big\}, \qquad p_{\mathbf{w}}(k) = 0 \ \text{ for } k < 0, \tag{2.1}$$

which counts the monomials of weighted degree k. A polynomial $f \in S_d$ is called $\mathbf{w}$-homogeneous of degree d; equivalently,

$$f(t^{w_1} z_1, \dots, t^{w_N} z_N) = t^d f(z) \qquad (t \in \mathbb{C}^*). \tag{2.2}$$

Its Jacobian ideal and graded *Milnor algebra* are, respectively,

$$\mathcal{J}_f = (\partial_1 f, \dots, \partial_N f), \qquad \mathbb{M}_f = S/\mathcal{J}_f = \bigoplus_{\ell \geq 0} (\mathbb{M}_f)_\ell. \tag{2.3}$$

Recall from Definition 1.1 that f is *admissible* if it has an isolated critical point at the origin; write $U_{\mathbf{w},d} \subseteq S_d$ for the set of such polynomials of degree d. By Lemma 3.1 below, f is admissible exactly when the common zero set of its partial derivatives is $\{0\}$. Euler's identity is

$$d\, f = \sum_i w_i z_i \partial_i f. \tag{2.4}$$

Thus every critical point of f lies on $\{f = 0\}$, so this means that the affine cone $\{f = 0\} \subset \mathbb{C}^N$ is smooth away from the origin, that is, that the hypersurface X_f is *quasi-smooth*. The pair $(\mathbf{w}, d)$ is called non-degenerate when $U_{\mathbf{w},d} \neq \emptyset$, and Calabi–Yau when moreover $d = \sum_i w_i$. The non-degenerate Calabi–Yau weight systems are classified for small n: there are 95 for $n = 2$ [Y1990, IF2000], 7555 for $n = 3$ [CLS1990, KS1994, KS1992a] and 1 100 055 for $n = 4$ [LSW1999], all available from [KS2026a, KS2026b]. The degree d is fixed throughout and is suppressed from the notation $P_{\mathbf{w}}$ and $\mu(\mathbf{w})$ below; in the Calabi–Yau case it is determined by $\mathbf{w}$, and $U_{\mathbf{w}} = U_{\mathbf{w},d}$. The following power series and its value at $t = 1$ will appear repeatedly:

$$P_{\mathbf{w}}(t) = \prod_{i=1}^N \frac{1 - t^{\,d - w_i}}{1 - t^{\,w_i}} = \sum_{k \geq 0} a_k t^k, \qquad \mu(\mathbf{w}) = \lim_{t \to 1} P_{\mathbf{w}}(t) = \prod_{i=1}^N \Big(\frac{d}{w_i} - 1 \Big). \tag{2.5}$$

The limit exists factor by factor, and Theorem 3.2 shows that $P_{\mathbf{w}}$ is a polynomial when $(\mathbf{w}, d)$ is non-degenerate. Expanding the numerator and the geometric series in the denominator expresses the coefficients, for every $(\mathbf{w}, d)$, as

$$a_\ell = \sum_{I \subseteq \{1, \dots, N\}} (-1)^{|I|}\, p_{\mathbf{w}}\Big(\ell - \sum_{i \in I} (d - w_i) \Big), \tag{2.6}$$

an identity of formal power series that involves no polynomial f. Finally, $\mu_r \subset \mathbb{C}^*$ denotes the group of r-th roots of unity, empty products and least common multiples equal 1, and Hodge numbers with an index outside the allowed range are 0.

2.2 Links and Sasakian Hodge numbers

For $f \in U_{\mathbf{w},d}$ let $V_f = f^{-1}(0) \subset \mathbb{C}^N$ and $Y = V_f \setminus 0$. The link $K_f = V_f \cap S^{2N-1}$ is a smooth closed manifold of dimension $2n + 1$, and it is $(n-1)$-connected [M1968]; its Betti numbers are $b_k(K_f) = \dim_{\mathbb{Q}} H^k(K_f; \mathbb{Q})$. The weighted action of $\mathbb{C}^*$ on $\mathbb{C}^N$ and its Euler vector field are

$$t \cdot z = (t^{w_1} z_1, \dots, t^{w_N} z_N), \qquad E_{\mathbf{w}} = \sum_i w_i z_i \partial_{z_i}. \tag{2.7}$$

The action preserves Y, its circle subgroup preserves K_f, and its infinitesimal action is generated by $E_{\mathbf{w}}$. The stabiliser of $z \in Y$ is $\mu_{m(z)}$ with $m(z) = \gcd\{w_i : z_i \neq 0\}$; since $\gcd(\mathbf{w}) = 1$ the generic stabiliser is trivial.

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The orbifold and its coarse space are distinguished throughout. The quotient stack $\mathcal{X}_f = [Y/\mathbb{C}^*]$ is a smooth proper Deligne–Mumford stack, that is, a compact complex orbifold, of dimension n. Coherent sheaves on $\mathcal{X}_f$ are $\mathbb{C}^*$-equivariant coherent sheaves on Y, $\Omega^p_{\mathcal{X}_f}$ denotes the sheaf of holomorphic p-forms on $\mathcal{X}_f$, and $\mathcal{O}_{\mathcal{X}_f}(k)$ denotes the line bundle whose sections are functions of weighted degree k. The coarse space $X_f = Y/\mathbb{C}^* \subset \mathbb{P}^{N-1}(\mathbf{w})$ is a projective variety with cyclic quotient singularities. The Hodge numbers of the orbifold are

$$h^{p,q}(\mathcal{X}_f) = \dim H^q(\mathcal{X}_f, \Omega^p_{\mathcal{X}_f}). \tag{2.8}$$

Since taking invariants of a finite group is exact in characteristic zero, coherent cohomology on $\mathcal{X}_f$ is the cohomology of the pushforward to the projective variety X_f, so analytic and algebraic coherent cohomology agree by GAGA. The pushforward of $\Omega^p_{\mathcal{X}_f}$ is Steenbrink's sheaf $\tilde{\Omega}^p_{X_f}$ of differential forms on the V-manifold X_f, and

$$H^k(X_f;\mathbb{C}) = \bigoplus_{p+q=k} H^q(X_f, \tilde{\Omega}^p_{X_f})$$

is a pure Hodge structure [S1977b, §1], [S1977a, §2]. Hence $h^{p,q}(\mathcal{X}_f)$ are the Hodge numbers of $H^{p+q}(X_f;\mathbb{Q})$, and the same holds for the closed suborbifolds of $\mathcal{X}_f$ that appear in Section 4.

A Sasakian structure on a manifold M of dimension $2n+1$ is a tuple (ξ, η, Φ, g) consisting of a contact form η with Reeb field ξ, an endomorphism Φ of TM and a Riemannian metric g, such that the metric cone $(M \times \mathbb{R}_{>0}, dr^2 + r^2 g)$ is Kähler with a complex structure that maps $r\partial_r$ to ξ and restricts to Φ on $\ker \eta$ [BG2008, Ch. 6]. It is *quasi-regular* if all orbits of ξ are closed; then ξ generates a locally free circle action, and the space of leaves is a compact Kähler orbifold. The weighted Sasakian structure of S^{2N-1}, whose Reeb field generates the weighted circle action, restricts to a quasi-regular Sasakian structure (ξ, η, Φ, g) on K_f [BG2008, Ch. 9]. Since the real weighted dilations move Y transversally to the sphere, its space of leaves is $\mathcal{X}_f$, and $K_f \to \mathcal{X}_f$ is a Seifert circle orbibundle. In what follows the structure is abbreviated to (ξ, η).

In the Calabi–Yau case $d = \sum_i w_i$ the orbifold $\mathcal{X}_f$ has a holomorphic volume form. The residue

$$\Omega_Y = \mathrm{Res}_{f=0}\big(dz_1 \wedge \cdots \wedge dz_N / f\big)$$

is a nowhere-vanishing holomorphic $(n+1,0)$-form on Y of weight $\sum_i w_i - d = 0$, so that $\mathcal{L}_{E_\mathbf{w}} \Omega_Y = 0$. Its contraction $\varepsilon = \iota_{E_\mathbf{w}} \Omega_Y$ is a holomorphic $(n,0)$-form with $\iota_{E_\mathbf{w}} \varepsilon = 0$, and it is closed by Cartan's formula:

$$d\varepsilon = \mathcal{L}_{E_\mathbf{w}} \Omega_Y - \iota_{E_\mathbf{w}} d\Omega_Y = 0.$$

Being invariant and horizontal, ε descends to a nowhere-vanishing section of the canonical bundle of $\mathcal{X}_f$, and on K_f it restricts to a closed basic $(n,0)$-form that is nowhere vanishing on $\ker \eta$. Following [TV2008, Def. 3.1], a *contact Calabi–Yau* structure is a Sasakian structure together with a closed basic $(n,0)$-form ε with

$$\varepsilon \wedge \bar{\varepsilon} = (-1)^{n(n+1)/2}\, i^n\, (d\eta)^n;$$

such a structure is null Sasakian and transversally Ricci-flat [TV2008, Cor. 3.7]. The structure restricted from the weighted sphere is null Sasakian, but its transverse metric is in general not Ricci-flat. By El Kacimi's transverse Calabi–Yau theorem [E1990] it can be deformed to $\eta' = \eta + \zeta$, with ζ a basic 1-form, so that the Reeb field, the transverse holomorphic structure and the basic class $[d\eta]$ are unchanged and the transverse metric is Ricci-flat [HV2015, Prop. 6.6, Prop. 6.7], [CRS2020, Prop. 13]. For the deformed structure $\log|\varepsilon|^2$ is a basic pluriharmonic function, hence constant, so after rescaling ε by a positive constant K_f is contact Calabi–Yau. The Hodge-theoretic invariants studied below depend only on the Reeb foliation, its transverse holomorphic structure and the class $[d\eta]$, so they are the same for both Sasakian structures.

A differential form α on K_f is *basic* if $\iota_\xi \alpha = 0$ and $\mathcal{L}_\xi \alpha = 0$. Basic forms make up a subcomplex of the de Rham complex, whose cohomology with real coefficients is the basic cohomology $H^*_B(K_f)$. The transverse complex structure splits complex-valued basic forms by type, and the resulting basic Dolbeault cohomology groups $H^{p,q}_B(K_f)$ have dimensions $h^{p,q}_B$. The form $d\eta$ is a closed basic $(1,1)$-form, the transverse Kähler form, and $L = [d\eta] \wedge \cdot$ denotes the corresponding Lefschetz operator. A class in H^k_B, or in $H^{p,q}_B$ with $p+q=k$, is *primitive* if $k \le n$ and it is annihilated by L^{n-k+1}.

Definition 2.1 (Sasakian Hodge numbers)**.** On a compact Sasakian $(2n+1)$-manifold M the harmonic k-forms with $k \le n$ are basic and primitive [T1965], and they split by type [T1975, I2004]. The *Sasakian*

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Hodge number $h_S^{p,q}(M)$, for $p+q \le n$, is the dimension of the space of harmonic forms of type (p,q). By basic Hodge theory for transversely Kähler foliations [E1990, Prop. 3.4.5, Thm. 3.4.6, §3.4.7], this space is isomorphic to the primitive part of $H_B^{p,q}(M)$, so that

$$h_S^{p,q}(M) = h_B^{p,q}(M) - h_B^{p-1,q-1}(M) \qquad (p+q \le n). \tag{2.9}$$

The following description underlies the computations of [AHHH+2024a].

Theorem 2.2 ([S1977a, I2004])**.** *Let f be a weighted homogeneous polynomial of degree d in $N = n+2$ variables, with an isolated critical point at the origin. Write K_f for its link and $\mathbb{M}_f = S/\mathcal{J}_f$ for its graded Milnor algebra, as in Section 2.1, and let $p+q=n$. Then*

$$h_S^{p,q}(K_f) = \dim(\mathbb{M}_f)_\ell, \qquad \ell = (p+1)d - \sum_i w_i. \tag{2.10}$$

In the Calabi–Yau case $\ell = pd$.

Itoh's proof identifies the harmonic forms on K_f with primitive harmonic forms on the orbifold $\mathcal{X}_f$ [I2004], and then appeals to the Jacobian-ring description of the primitive cohomology of quasi-smooth hypersurfaces [G1969a, G1969b, S1977a, D1982]. In the convention of [D1982, Thm. 4.3.2] the space $H^{p,q}_{\mathrm{prim}}(X_f)$ corresponds to the degree $(q+1)d - \sum_i w_i$, with p and q exchanged; both conventions give the same numbers by Hodge symmetry. The alternative derivation in Theorem B is proved in Section 3.3.

For 7-dimensional Calabi–Yau links, that is, for $N=5$ and $d = \sum_i w_i$, the Crowley–Nordström invariant $\nu \in \mathbb{Z}/48$ of the G_2-structure $\varphi = \eta' \wedge d\eta' + \operatorname{Im}\varepsilon$ induced by the contact Calabi–Yau structure (η', ε) above [CRS2020, Prop. 11] was also computed in [AHHH+2024a]. For a closed 7-manifold M with a G_2-structure φ it is defined by [CN2015]

$$\nu(\varphi) = \chi(W) - 3\sigma(W) \bmod 48. \tag{2.11}$$

Here W is any compact 8-manifold with boundary M, oriented compatibly with the boundary orientation of M, that carries a Spin(7)-structure inducing φ; this is a reduction of the structure group of TW, and no torsion-free metric is required. The numbers $\chi(W)$ and $\sigma(W)$ are the Euler characteristic and the signature of W. For a link, W may be taken to be the compact Milnor fibre $f^{-1}(\delta) \cap B^{10}$, with the orientation of a complex manifold and the Spin(7)-structure induced by a holomorphic volume form [CRS2020, §3]. It has the homotopy type of a bouquet of $\mu(\mathbf{w})$ four-spheres [M1968, MO1970], so $\chi(W) = 1 + \mu(\mathbf{w})$. If $\{z^\alpha\}$ is a monomial basis of $\mathbb{M}_f$ and $l(\alpha) = \sum_i (\alpha_i+1)w_i/d$, Steenbrink's signature formula [S1977a] gives $\sigma(W) = \mu_+ - \mu_-$, and hence [CRS2020, Thm. 2]

$$\nu = 1 + \mu - 3(\mu_+ - \mu_-) \bmod 48, \qquad \mu_\pm = \#\big\{\alpha :\ l(\alpha) \notin \mathbb{Z},\ \lfloor l(\alpha) \rfloor \text{ even } (+) \text{ or odd } (-)\big\}. \tag{2.12}$$

2.3 Statement of the conjectures

The conjectures of [AHHH+2024a] are recalled here in the present notation. Two admissible polynomials f and g are *R-equivalent* if $g = f \circ \phi$ for a germ of biholomorphism $\phi : (\mathbb{C}^N, 0) \to (\mathbb{C}^N, 0)$; then ϕ^* maps $\mathcal{J}_f$ to $\mathcal{J}_g$, so their Milnor algebras are isomorphic as algebras. Two $\mathbf{w}$-homogeneous polynomials f, g of degree d are called *weakly R-equivalent* in [AHHH+2024a] if $(\mathbb{M}_f)_\ell$ and $(\mathbb{M}_g)_\ell$ have the same dimension for each $\ell = (p+1)d - \sum_i w_i$ with $p+q=n$, that is, in each degree that appears in (2.10).

Weight Determinacy Conjecture (Conjecture 7 of [AHHH+2024a]). *If two weighted homogeneous polynomials f and g of the same degree d have the same weight vector up to permutation, then f and g are weakly R-equivalent.*

The surrounding discussion in [AHHH+2024a] restricts to polynomials with "no further singularities" than the one at the origin, and adds that the Sasakian Hodge numbers and the Crowley–Nordström invariants then agree.

Calabi–Yau Bound Conjecture (Conjecture 8 of [AHHH+2024a]). *The Sasakian Hodge number $h_S^{2,1}$ of a 7-dimensional Calabi–Yau link is bounded above by the Hodge number $h_{CY}^{2,1}$ of the Calabi–Yau threefold built from the same weighted homogeneous polynomial: $h_S^{2,1} \le h_{CY}^{2,1}$.*

The Calabi–Yau Bound Conjecture is numbered Conjecture 11 in the first arXiv version of [AHHH+2024a]. The values $h_{CY}^{2,1}$ used there are those of the Kreuzer–Skarke list [KS2026a].

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3 Proof of the Weight Determinacy Conjecture

We first compute the Hilbert series of the Milnor algebra $\mathbb{M}_f = S/\mathcal{J}_f$ for an admissible polynomial $f \in U_{\mathbf{w},d}$. Together with Theorem 2.2, this proves weight determinacy and gives explicit formulae in the denumerants $p_{\mathbf{w}}$ of (2.1). We then derive the same Hodge numbers directly from sheaf cohomology on $\mathcal{X}_f$, before applying the formulae to the six outstanding threefold weight systems.

3.1 The Hilbert series of the Milnor algebra

The proof rests on two elementary facts: an admissible polynomial has no critical points outside the origin, and its partial derivatives form a homogeneous system of parameters in the Cohen–Macaulay ring S. The first lemma supplies the isolatedness and positivity needed for that argument.

Lemma 3.1. *Let $f \in \mathbb{C}[z_1, \dots, z_N]$ be weighted homogeneous of degree d for positive weights $\mathbf{w}$, with an isolated critical point at the origin; equivalently, $f \in U_{\mathbf{w},d}$ in the notation of Section 2.1. Then the common zero set of $\partial_1 f, \dots, \partial_N f$ in $\mathbb{C}^N$ is $\{0\}$, and $d > w_i$ for every i.*

Proof. Since $\partial_i f(t \cdot z) = t^{d-w_i}\partial_i f(z)$, the critical set of f is invariant under the $\mathbb{C}^*$-action. All weights are positive, so the closure of every orbit contains the origin. A critical point $z \neq 0$ would therefore produce critical points $t \cdot z$ accumulating at 0 as $t \to 0$, which contradicts isolatedness. If $d \leq w_i$ for some i, the only monomial of degree at most d involving z_i is z_i itself, and since f has a critical point at 0 it does not contain z_i; but then the whole z_i-axis is critical, again a contradiction. □

Theorem 3.2. *Let $S = \mathbb{C}[z_1, \dots, z_N]$ have grading $\deg z_i = w_i$, and let $f \in S_d$ have an isolated critical point at the origin. Write $\mathbb{M}_f = S/(\partial_1 f, \dots, \partial_N f)$, and let $P_{\mathbf{w}}$, a_ℓ and $\mu(\mathbf{w})$ be as in (2.5). Then $\partial_1 f, \dots, \partial_N f$ form a regular sequence in S, and*

$$\sum_{\ell \geq 0} \dim(\mathbb{M}_f)_\ell \, t^\ell = P_{\mathbf{w}}(t) = \prod_{i=1}^{N} \frac{1 - t^{\,d-w_i}}{1 - t^{\,w_i}} \,.$$

In particular $P_{\mathbf{w}}$ is a polynomial, $\dim(\mathbb{M}_f)_\ell = a_\ell$ depends only on $(\mathbf{w}, d)$ for every ℓ, and $\dim \mathbb{M}_f = \mu(\mathbf{w})$.

Proof. By Lemma 3.1 and the Nullstellensatz, the radical of $\mathcal{J}_f$ is the maximal ideal $\mathfrak{m} = (z_1, \dots, z_N)$. Hence the N homogeneous generators $\partial_i f$, of positive degrees $d - w_i$, form a homogeneous system of parameters of the graded ring S. Since S is Cohen–Macaulay, they form a regular sequence [S1978, §3], [BH1993, §2.1]. The Koszul complex of a regular sequence is exact, so it is a graded free resolution

$$0 \to S\Big(-\sum_i (d-w_i)\Big) \to \cdots \to \bigoplus_{i<j} S(-(d-w_i)-(d-w_j)) \to \bigoplus_i S(-(d-w_i)) \to S \to \mathbb{M}_f \to 0.$$

Taking the alternating sum of Hilbert series, and using $\sum_k \dim S_k \, t^k = \prod_i (1 - t^{w_i})^{-1}$, gives the stated product. Its value at $t = 1$ is $\prod_i (d - w_i)/w_i = \mu(\mathbf{w})$. □

The identity of Theorem 3.2 is classical, and is implicit in the computation of the Milnor number $\mu(\mathbf{w})$ by Milnor and Orlik [MO1970]; see also [S1978, S2026] for the algebraic argument and [KS1992a, eq. (2)] for its use in the classification of Calabi–Yau weight systems. Theorem A follows at once, proving the Weight Determinacy Conjecture in every graded degree and answering Itoh's weight-dependence question [I2004, p. 123].

Proof of Theorem A. A permutation of the coordinates identifies $\mathbb{M}_g$ with the Milnor algebra of a polynomial with the same weights as f, and Theorem 3.2 applies to both. The Sasakian Hodge numbers are dimensions of graded pieces by Theorem 2.2. For (2.12), a monomial z^α of degree k has $l(\alpha) = (k + \sum_i w_i)/d$, which depends only on k, so μ_+ and μ_- are determined by the Hilbert function. □

The isolated-singularity hypothesis is part of the setting of [AHHH+2024a], but it is not contained in the displayed statement of the Weight Determinacy Conjecture, and it cannot be omitted.

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Example 3.3 (the hypothesis is necessary)**.** Take $\mathbf{w} = (1,1,1,1,1)$ and $d = 5$. The Fermat quintic $f = \sum_{i=1}^{5} z_i^5$ is admissible, $\mathcal{J}_f = (z_1^4, \dots, z_5^4)$ and $\dim(\mathbb{M}_f)_{10} = 101$. The polynomial $g = \sum_{i=1}^{4} z_i^5$ has the same weights and degree but is singular along the z_5-axis; here $\mathcal{J}_g = (z_1^4, \dots, z_4^4)$, $\dim(\mathbb{M}_g)_{10} = 251$, and $\mathbb{M}_g$ is infinite-dimensional.

The conclusion of Theorem A concerns dimensions only, and this is as much as can be expected: the Milnor algebras themselves are in general not isomorphic, so weak R-equivalence is strictly weaker than R-equivalence [A2012, MY1982].

Example 3.4 (the Milnor algebras differ)**.** In five variables, let $F = \sum_i z_i^5$ be the Fermat quintic and let $f_\psi = \sum_i z_i^5 - 5\psi z_1 \cdots z_5$ be a Dwork quintic, with $\psi \neq 0$ and $\psi^5 \neq 1$. By Theorem 3.2 the two Milnor algebras have the same Hilbert function. In $\mathbb{M}_F$ the linear form z_1 satisfies $z_1^4 = 0$. Suppose that some $x \in \mathbb{M}_{f_\psi}$ of degree one satisfied $x^4 = 0$, and write $x = \sum_i a_i z_i \neq 0$. Then x^4 would lie in

$$(\mathcal{J}_{f_\psi})_4 = \operatorname{span}\{z_i^4 - \psi \prod_{j \neq i} z_j\}.$$

The coefficient of $z_i^3 z_j$ in x^4 is $4a_i^3 a_j$, so at most one a_i is nonzero; but then $x^4 = a_i^4 z_i^4$, which is not in that span since $\psi \neq 0$. Hence $\mathbb{M}_F \ncong \mathbb{M}_{f_\psi}$ as graded algebras. Both algebras are generated in degree one, so an isomorphism of local algebras would induce an isomorphism of their associated graded algebras with respect to the maximal ideal, which are $\mathbb{M}_F$ and $\mathbb{M}_{f_\psi}$ themselves. Hence the Milnor algebras are not isomorphic even as ungraded algebras, and F and f_ψ are not R-equivalent.

Remark 3.5 (the Crowley–Nordström invariant)**.** The statement of Theorem A about ν concerns only $N = 5$ and $d = \sum_i w_i$, where the link carries the G_2-structure of Section 2.2. For every admissible f the invariant ν of that structure is given by (2.12), whose ingredients $\chi(W) = 1 + \mu(\mathbf{w})$ and $\sigma(W) = \mu_+ - \mu_-$ are determined by the Hilbert function of $\mathbb{M}_f$; by Theorem A, ν therefore depends only on $\mathbf{w}$. The rest of Theorem A holds for every non-degenerate $(\mathbf{w}, d)$, including links that carry no such G_2-structure.

3.2 Closed formulae

Theorem 3.2 reduces the Sasakian Hodge numbers to coefficients of $P_{\mathbf{w}}$, which are given by the short alternating sums (2.6) of denumerants.

Corollary 3.6. *Let f be a weighted homogeneous polynomial of degree d with an isolated critical point at the origin, in $N = n + 2$ variables of weights $\mathbf{w}$. Let $a_k = \dim(\mathbb{M}_f)_k$ be the Hilbert coefficients of Theorem 3.2, and let $p_{\mathbf{w}}$ denote the monomial count (2.1). Then $\mathbb{M}_f$ is a graded Artinian Gorenstein algebra with socle in degree $s = \sum_i (d - 2w_i)$, so $a_k = a_{s-k}$. In the Calabi–Yau case $s = nd$, and $h_S^{p,q}(K_f) = a_{pd} = a_{qd}$ for $p + q = n$. Moreover, with $m = \min(p, q)$, the coefficient a_{md} is the sum of the terms of (2.6) with $|I| \leq m$, all other terms being zero. Explicitly,*

$$\begin{aligned}
&n = 2: \quad h_S^{2,0} = 1, \quad h_S^{1,1} = p_{\mathbf{w}}(d) - \textstyle\sum_i p_{\mathbf{w}}(w_i);\\
&n = 3: \quad h_S^{3,0} = 1, \quad h_S^{2,1} = p_{\mathbf{w}}(d) - \textstyle\sum_i p_{\mathbf{w}}(w_i);\\
&n = 4: \quad h_S^{4,0} = 1, \quad h_S^{3,1} = p_{\mathbf{w}}(d) - \textstyle\sum_i p_{\mathbf{w}}(w_i), \quad h_S^{2,2} = p_{\mathbf{w}}(2d) - \textstyle\sum_i p_{\mathbf{w}}(d + w_i) + \sum_{i<j} p_{\mathbf{w}}(w_i + w_j).
\end{aligned}$$

Proof. By Theorem 3.2, $\mathbb{M}_f$ is a graded Artinian complete intersection. Such an algebra is Gorenstein, and its Hilbert function is symmetric [S1978, Thm. 4.1]; directly, $t^s P_{\mathbf{w}}(1/t) = P_{\mathbf{w}}(t)$. In the Calabi–Yau case, for $\ell = md$ the argument of $p_{\mathbf{w}}$ in (2.6) is $(m - |I|)d + \sum_{i \in I} w_i$. If $|I| = m + 1 \leq N - 1$ this is negative because $\sum_{i \in I} w_i < d$, and if $|I| \geq m + 2$ it is at most $-d$. Taking $m = \min(p, q)$, which is allowed since $a_{pd} = a_{qd}$, gives the stated truncation, and Theorem 2.2 identifies a_{pd} with $h_S^{p,q}$. □

The truncation holds in the degree md with $m = \min(p, q)$, but not in the symmetric degree. For the quintic,

$$h_S^{2,1} = a_5 = p_{\mathbf{w}}(5) - 5p_{\mathbf{w}}(1) = 126 - 25 = 101,$$

whereas in degree 10 the terms with $|I| = 2$ do not vanish:

$$\begin{aligned}
a_{10} &= p_{\mathbf{w}}(10) - 5p_{\mathbf{w}}(6) + 10p_{\mathbf{w}}(2)\\
&= 1001 - 1050 + 150 = 101.
\end{aligned}$$

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Remark 3.7 (polynomial deformations)**.** In the Calabi–Yau case the number $a_d = p_{\mathbf{w}}(d) - \sum_i p_{\mathbf{w}}(w_i)$ has a deformation-theoretic meaning. The group G of weighted coordinate changes $z_i \mapsto g_i(z)$, with $g_i \in S_{w_i}$, has Lie algebra $\bigoplus_i S_{w_i}$, and the differential of its orbit map at f is

$$\bigoplus_i S_{w_i} \longrightarrow S_d, \qquad (g_i) \longmapsto \sum_i g_i\, \partial_i f,$$

with image $(\mathcal{J}_f)_d$. This map is injective by the regular-sequence statement of Theorem 3.2: a relation $\sum_i g_i \partial_i f = 0$ is a syzygy of the regular sequence $\partial_1 f, \dots, \partial_N f$, hence a combination of Koszul syzygies, whose coefficients would have degree $w_i + w_j - d < 0$. By Euler's identity $f \in (\mathcal{J}_f)_d$, so the rescaling of the equation is already contained in the image. Hence $a_d = \dim S_d/(\mathcal{J}_f)_d$ is the dimension of the space of first-order deformations of f modulo weighted coordinate changes and rescaling, that is, the number of polynomial deformations of X_f; equivalently, since the stabiliser of f in G is finite, the G-orbit of f has codimension a_d in S_d. For the quintic it is $126 - 25 = 101$, and for the example $\mathbf{w} = (22, 29, 49, 50, 75)$ of [AHHH+2024a] it is $7 - 5 = 2$. For K3 surfaces it counts deformations of X_f as a hypersurface in $\mathbb{P}(\mathbf{w})$, not the 20-dimensional deformation space of an unpolarised K3 surface. The Gröbner-basis approach of [AHHH+2024a] obtains the same numbers after enumerating a monomial basis of the whole Milnor algebra, of size $\mu(\mathbf{w})$, whereas Corollary 3.6 needs $\sum_{j=0}^{m} \binom{N}{j}$ denumerants for $h_S^{p,q}$ with $m = \min(p, q)$. This is $N + 1$ for $h_S^{n-1,1}$ in every dimension, and at most $1 + N + \binom{N}{2}$, which is 22 for $N = 6$, for every Hodge number in dimensions $n \le 5$.

3.3 A sheaf-cohomological proof

We prove Theorem B by computing the Hodge numbers directly on the quotient orbifold. The five lemmas below organise the argument into three stages. First, Lemma 3.8 identifies basic cohomology with coherent cohomology on the leaf space. Next, the Gysin sequence of Lemma 3.9 and the connectivity calculation of Lemma 3.10 isolate the primitive contribution in the middle degree. Finally, Lemmas 3.11 and 3.12 compute the Euler characteristics from which the remaining Hodge numbers are recovered. Combining the resulting formula with Theorem 3.2 recovers Itoh's formula, as explained in Section 1.

The proof uses the notation of Section 2.2, with $\mathcal{X} = \mathcal{X}_f$. Let $A = S/(f)$, a graded ring with Hilbert function $h_A(k) = \dim A_k = p_{\mathbf{w}}(k) - p_{\mathbf{w}}(k - d)$, and put $\kappa = d - \sum_i w_i$. Lemmas 3.11 and 3.12 below do not involve the link: they hold for every $f \in U_{\mathbf{w},d}$ in $N = r + 2 \ge 3$ variables, with r in place of n, and they are used in this generality in Section 4.

We begin by identifying the basic Dolbeault complex with the Dolbeault complex of the quotient orbifold. This makes it possible to compute the basic Hodge numbers by coherent sheaf cohomology.

Lemma 3.8 (basic forms are forms on the leaf space)**.** *Let K_f be the quasi-regular Sasakian link of an admissible weighted homogeneous polynomial $f \in U_{\mathbf{w},d}$, and let $\mathcal{X} = \mathcal{X}_f$ be its quotient orbifold as in Section 2.2. For all p, q there are isomorphisms $H_B^{p,q}(K_f) \cong H^q(\mathcal{X}, \Omega_{\mathcal{X}}^p)$; in particular $h_B^{p,q}(K_f) = h^{p,q}(\mathcal{X})$.*

Proof. The weighted radial flow identifies Y circle-equivariantly with $K_f \times \mathbb{R}_{>0}$ as a smooth manifold, since the real weighted dilations $z \mapsto e^r \cdot z$ move Y transversally to the sphere. Basic forms on K_f therefore correspond to forms on Y that are invariant under $\mathbb{C}^*$ and annihilated by the real and imaginary parts of the Euler vector field $E_{\mathbf{w}}$, which are exactly the differential forms on $\mathcal{X}$. The transverse complex structure of the Reeb foliation is the one induced from Y [BG2008, Ch. 9], so this correspondence respects types and $\bar\partial$. In a uniformising chart the $\bar\partial$-Poincaré lemma holds for invariant forms, by averaging over the finite stabiliser, so smooth basic $(p, \bullet)$-forms form a fine resolution of $\Omega_{\mathcal{X}}^p$; compare [GNT2016, §1.2.2]. □

To extract the primitive contribution, we next compare basic cohomology with the ordinary cohomology of the link. The connecting map is the Lefschetz operator introduced in Section 2.2.

Lemma 3.9 (Gysin sequence)**.** *Let K_f be the Sasakian link of $f \in U_{\mathbf{w},d}$, with contact form η, and put $L = [d\eta] \wedge \cdot$ on its real basic cohomology. There is a long exact sequence*

$$\cdots \to H_B^{k-2}(K_f) \xrightarrow{L} H_B^k(K_f) \to H^k(K_f; \mathbb{R}) \to H_B^{k-1}(K_f) \xrightarrow{L} H_B^{k+1}(K_f) \to \cdots$$

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Proof. Averaging over the circle shows that circle-invariant forms compute $H^*(K_f;\mathbb{R})$. Every invariant form decomposes uniquely as $\gamma = \alpha + \eta\wedge\beta$ with α,β basic, namely $\beta = \iota_\xi\gamma$, and $d\gamma = d\alpha + d\eta\wedge\beta - \eta\wedge d\beta$. Hence $\gamma \mapsto \iota_\xi\gamma$ is, up to sign, a surjective chain map onto the basic complex shifted by one, with kernel the basic complex. The connecting homomorphism of the resulting short exact sequence of complexes sends $[\beta]$ to $[d(\eta\wedge\beta)] = [d\eta\wedge\beta]$. □

Milnor's connectivity theorem now determines all basic cohomology groups away from the middle degree. This is the reason that the Euler characteristics computed below suffice to recover the remaining Hodge numbers.

Lemma 3.10 (basic cohomology away from the middle)**.** *Let K_f be the $(2n+1)$-dimensional Sasakian link of an admissible polynomial $f \in U_{\mathbf{w},d}$, where $n \geq 2$, with contact form η and Lefschetz operator L as in Lemma 3.9. For $0 \leq k \leq 2n$ with $k \neq n$, the group $H^k_B(K_f)$ is spanned by $[d\eta]^{k/2}$ if k is even and vanishes if k is odd. Consequently $h^{p,q}_B = \delta_{pq}$ whenever $p+q \neq n$, the map $L : H^{k-2}_B \to H^k_B$ is injective for $k \leq n+1$, and $H^n(K_f;\mathbb{R}) \cong H^n_B / L\,H^{n-2}_B$. In particular,*

$$b_n(K_f) = \sum_{p+q=n} h^{p,q}_S(K_f). \tag{3.1}$$

Proof. Since K_f is $(n-1)$-connected [M1968], Poincaré duality gives $H^k(K_f;\mathbb{R}) = 0$ unless $k \in \{0, n, n+1, 2n+1\}$. Inserting this into Lemma 3.9: for $1 \leq k \leq n-1$ the map $L : H^{k-2}_B \to H^k_B$ is surjective and $L : H^{k-1}_B \to H^{k+1}_B$ is injective, so

$$H^1_B = 0, \qquad H^k_B = L^{\lfloor k/2\rfloor} H^{k-2\lfloor k/2\rfloor}_B \quad (k \leq n-1),$$

while L is injective on H^j_B for $j \leq n-2$. Basic forms of degree above $2n$ vanish, so the end of the sequence gives $H^{2n}_B \cong H^{2n+1}(K_f;\mathbb{R}) \cong \mathbb{R}$. For $n+2 \leq k \leq 2n$ the same argument shows that $L : H^j_B \to H^{j+2}_B$ is bijective for $n+1 \leq j \leq 2n-2$ and that H^{2n-1}_B injects into $H^{2n+1}_B = 0$. This determines the dimensions as claimed. The class $[d\eta]^n$ is nonzero: if $(d\eta)^n = d\beta$ with β basic, then Stokes' theorem gives

$$\int_{K_f} \eta\wedge(d\eta)^n = \int_{K_f} d\eta\wedge\beta = 0,$$

since $d\eta\wedge\beta$ is a basic form of degree $2n+1$; but $\eta\wedge(d\eta)^n$ is a volume form. Hence every $[d\eta]^j$ is nonzero, and it spans H^{2j}_B when $2j \neq n$. The basic Hodge decomposition [E1990, Thm. 3.4.6] is

$$H^k_B \otimes \mathbb{C} = \bigoplus_{p+q=k} H^{p,q}_B.$$

Since $d\eta$ has type $(1,1)$, this gives $h^{p,q}_B = \delta_{pq}$ for $p+q \neq n$. Finally, $L : H^{n-1}_B \to H^{n+1}_B$ is injective: for n even its source vanishes, and for n odd its source is spanned by $[d\eta]^{(n-1)/2}$, whose image $[d\eta]^{(n+1)/2}$ is nonzero. The exact sequence of Lemma 3.9 at $k=n$ therefore gives $H^n(K_f;\mathbb{R}) \cong H^n_B / L\,H^{n-2}_B$. Here $L\,H^{n-2}_B$ is spanned by $[d\eta]^{n/2}$ for n even and vanishes for n odd, so $b_n(K_f) = \sum_{p+q=n}(h^{p,q}_B - \delta_{pq})$, which is $\sum_{p+q=n} h^{p,q}_S$ by (2.9). □

The remaining calculation takes place on the quotient orbifold. We state the next two lemmas also for curves, since the twisted loci in Section 4 can have complex dimension one.

Lemma 3.11 (line bundles on the leaf space)**.** *Let f be weighted homogeneous of degree d with an isolated critical point at the origin, in $N = r+2 \geq 3$ variables of weights $\mathbf{w}$, and let $\mathcal{X} = [(f^{-1}(0)\setminus 0)/\mathbb{C}^*]$. Set $A = \mathbb{C}[z_1,\dots,z_N]/(f)$, $h_A(k) = \dim A_k$ and $\kappa = d - \sum_i w_i$. For every $k \in \mathbb{Z}$,*

$$\begin{aligned} H^0(\mathcal{X},\mathcal{O}_\mathcal{X}(k)) &\cong A_k, \\ H^i(\mathcal{X},\mathcal{O}_\mathcal{X}(k)) &= 0 \qquad (0 < i < r), \\ \dim H^r(\mathcal{X},\mathcal{O}_\mathcal{X}(k)) &= h_A(\kappa - k). \end{aligned} \tag{3.2}$$

Hence

$$\chi\big(\mathcal{X},\mathcal{O}_\mathcal{X}(k)\big) = h_A(k) + (-1)^r h_A(\kappa - k).$$

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Proof. Since $\mathbb{C}^*$ is linearly reductive, $H^i(\mathcal{X},\mathcal{O}_\mathcal{X}(k))$ is the weight-k part of $H^i(Y,\mathcal{O}_Y)$, and Y is the punctured spectrum of A. The ring A is a hypersurface ring of dimension $r+1\geq 2$, hence Cohen–Macaulay, so $H^0(Y,\mathcal{O}_Y)=A$ and $H^i(Y,\mathcal{O}_Y)\cong H^{i+1}_\mathfrak{m}(A)$ for $i\geq 1$, which vanishes unless $i=r$ [BH1993, Ch. 3]. The sequence

$$0\to S(-d)\xrightarrow{f} S\to A\to 0,$$

together with $H^j_\mathfrak{m}(S)=0$ for $j\neq N$ and $H^N_\mathfrak{m}(A)=0$, identifies $H^{r+1}_\mathfrak{m}(A)$ with the kernel of the surjection $f: H^N_\mathfrak{m}(S)(-d)\to H^N_\mathfrak{m}(S)$. The module $H^N_\mathfrak{m}(S)$ has a basis of inverse monomials $z^{-\beta}$ with all $\beta_i\geq 1$, so its degree-k part has dimension $p_\mathbf{w}(-k-\sum_i w_i)$. Hence

$$\begin{aligned}\dim H^{r+1}_\mathfrak{m}(A)_k &= p_\mathbf{w}(d-k-\sum_i w_i)-p_\mathbf{w}(-k-\sum_i w_i)\\ &= h_A(\kappa-k).\end{aligned}$$

□

The conormal and Euler sequences express the classes of differential forms in terms of the line bundles covered by Lemma 3.11. Only this identity of classes, together with the canonical bundle, is needed in the proof of Theorem B.

Lemma 3.12 (differential forms on the leaf space)**.** *Let $\mathcal{X}$ be the r-dimensional quotient orbifold of the quasi-smooth hypersurface in Lemma 3.11, with $\kappa=d-\sum_i w_i$, and write $\mathcal{E}=\bigoplus_{i=1}^N\mathcal{O}_\mathcal{X}(-w_i)$. In the Grothendieck group of vector bundles on $\mathcal{X}$,*

$$[\Omega^p_\mathcal{X}]=\sum_{i+j+m=p}(-1)^{i+j}\,[\mathcal{O}_\mathcal{X}(-jd)]\cdot\big[\textstyle\bigwedge^m\mathcal{E}\big],\qquad\text{and}\qquad \Omega^r_\mathcal{X}\cong\mathcal{O}_\mathcal{X}(\kappa).$$

Proof. The $\mathbb{C}^*$-equivariant vector bundles on Y below descend to $\mathcal{X}$. The conormal sequence of $Y\subset\mathbb{C}^N\setminus 0$,

$$0\to\mathcal{O}_Y(-d)\xrightarrow{df}\bigoplus_{i=1}^N\mathcal{O}_Y(-w_i)\,dz_i\to\Omega^1_Y\to 0,$$

is exact on the left because $df\neq 0$ on Y; here dz_i has weight w_i and df has weight d. It descends to an exact sequence

$$0\to\mathcal{O}_\mathcal{X}(-d)\to\mathcal{E}\to\mathcal{B}\to 0, \tag{3.3}$$

where $\mathcal{B}$ is the bundle on $\mathcal{X}$ obtained from Ω^1_Y. The pullbacks of forms on $\mathcal{X}$ are the forms on Y annihilated by the nowhere-vanishing vector field $E_\mathbf{w}$, so contraction with $E_\mathbf{w}$ gives a second exact sequence

$$0\to\Omega^1_\mathcal{X}\to\mathcal{B}\to\mathcal{O}_\mathcal{X}\to 0. \tag{3.4}$$

In terms of $\lambda_u(V)=\sum_m[\bigwedge^m V]u^m$, which is multiplicative in short exact sequences, the sequences (3.3) and (3.4) give

$$\lambda_u(\Omega^1_\mathcal{X})=\frac{\lambda_u(\mathcal{E})}{(1+u)(1+[\mathcal{O}_\mathcal{X}(-d)]u)}=\frac{\prod_i\big(1+[\mathcal{O}_\mathcal{X}(-w_i)]u\big)}{(1+u)(1+[\mathcal{O}_\mathcal{X}(-d)]u)},$$

whose coefficient of u^p is the stated identity. Taking determinants gives

$$\Omega^r_\mathcal{X}\cong\det\mathcal{B}\cong\det\mathcal{E}\otimes\mathcal{O}_\mathcal{X}(d)=\mathcal{O}_\mathcal{X}(d-\sum_i w_i).$$

□

Proof of Theorem B. By Definition 2.1 and Lemma 3.10, $h_S^{0,0}=1$, $h_S^{p,q}=0$ for $0<p+q<n$, and $h_S^{p,n-p}=h_B^{p,n-p}-\delta_{2p,n}$. Lemmas 3.8 and 3.10 then give

$$\chi(\mathcal{X},\Omega^p_\mathcal{X})=\sum_q(-1)^q h_B^{p,q}=(-1)^p+(-1)^{n-p}\,h_S^{p,n-p}.$$

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On the other hand, by Lemmas 3.12 and 3.11 with $r = n$,

$$\chi(\mathcal{X}, \Omega^p_{\mathcal{X}}) = \sum_{m=0}^{p} (-1)^{p-m} \sum_{|I|=m} \sum_{j=0}^{p-m} \chi\Big(\mathcal{O}_{\mathcal{X}}\big(-jd - \textstyle\sum_{i\in I} w_i\big)\Big).$$

The twist is negative except for $(I, j) = (\emptyset, 0)$, so each term equals $(-1)^n h_A(\kappa + jd + w_I)$, with $w_I = \sum_{i\in I} w_i$, apart from the extra summand 1 coming from $h_A(0)$. Comparing the two expressions,

$$h^{p,n-p}_S = \sum_{m=0}^{p} (-1)^m \sum_{|I|=m} \sum_{j=0}^{p-m} h_A(\kappa + jd + w_I).$$

Since $h_A(k) = p_{\mathbf{w}}(k) - p_{\mathbf{w}}(k-d)$, the sum over j telescopes to

$$p_{\mathbf{w}}(\kappa + (p-m)d + w_I) - p_{\mathbf{w}}(w_I - \sum_i w_i),$$

and the last term vanishes because $|I| \le p < N$. Since

$$\kappa + (p - |I|)d + w_I = (p+1)d - \sum_i w_i - \sum_{i\in I}(d - w_i),$$

this is the first equality in (1.2). For the second, in the formal expansion (2.6) with $\ell = (p+1)d - \sum_i w_i$ the terms with $|I| \ge p+1$ have argument $(p+1-|I|)d - \sum_{i\notin I} w_i < 0$, so the two sums coincide; this step does not use Theorem 3.2. □

Remark 3.13. The proof of Theorem B uses Milnor's connectivity theorem in Lemma 3.10, basic Hodge theory in Definition 2.1, and local cohomology of hypersurface rings in Lemma 3.11. It also produces the index convention of Theorem 2.2 directly, and the case $p = n$ reproduces the familiar count $h^{n,0} = p_{\mathbf{w}}(d - \sum_i w_i)$ of holomorphic volume forms. Term-by-term evaluations of (1.2) agree with classical examples, Hilbert-series coefficients, independent Gröbner-basis computations and the datasets below. The checks, including their scope in dimension one, are recorded in Appendix A under *Checks of the sheaf-cohomological formula.*

Remark 3.14 (the Sasakian Hodge diamond)**.** Theorem B determines every Sasakian Hodge number of K_f, but all except those in the middle degree take trivial values. Below the middle degree the link has no real cohomology, since it is $(n-1)$-connected [M1968], and the only basic classes there are the powers $[d\eta]^j$ of the transverse Kähler class (Lemma 3.10). For $j \ge 1$ the class $[d\eta]^j$ is not primitive, because $L^{n-2j+1}[d\eta]^j = [d\eta]^{n-j+1}$ is nonzero, so these classes contribute nothing to $h^{p,q}_S$, while the constant functions give $h^{0,0}_S = 1$. Above the middle degree no new numbers arise: the Hodge star maps the harmonic k-forms on K_f isomorphically onto the harmonic $(2n+1-k)$-forms, which is why Definition 2.1 only concerns $p+q \le n$. In the middle degree $h^{p,q}_S = h^{q,p}_S$ by complex conjugation, and in the Calabi–Yau case $h^{n,0}_S = p_{\mathbf{w}}(0) = 1$. The Sasakian Hodge numbers in degrees $k \le n$ therefore consist of $h^{0,0}_S = 1$, zeros in every degree $0 < k < n$, and the middle row $(1, h^{n-1,1}_S, \dots, h^{1,n-1}_S, 1)$; for a 9-dimensional link, for instance, $h^{1,0}_S$, $h^{1,1}_S$, $h^{2,1}_S$ and every other $h^{p,q}_S$ with $0 < p+q < 4$ vanish. In the Calabi–Yau case the only non-trivial numbers are thus $h^{1,1}_S$ for $n = 2$, $h^{2,1}_S$ for $n = 3$ and the pair $(h^{3,1}_S, h^{2,2}_S)$ for $n = 4$, and these are the Sasakian Hodge numbers listed in Sections 3.4, 5 and 6. They determine the real cohomology of the link:

$$b_0 = b_{2n+1} = 1, \qquad b_n = b_{n+1} = \sum_{p+q=n} h^{p,q}_S, \tag{3.5}$$

and all other Betti numbers vanish.

Remark 3.15 (moduli of transverse Calabi–Yau structures)**.** For a compact null-Sasakian manifold with fixed Reeb foliation, Moriyama showed that the moduli space $\mathfrak{M}_{CY}$ of transverse Calabi–Yau structures is a smooth manifold with tangent space [M2011, Thm. 6.5]

$$H^{n,0}_B \oplus H^{n-1,1}_B \oplus P^{1,1}_{B,\mathbb{R}}.$$

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The first two summands parametrise the transverse complex structures together with the holomorphic volume form, and the last, the real primitive basic $(1,1)$-classes, parametrises the transverse Kähler forms; for $n=3$ the dimension is $b_3 + b_1 + h_B^{1,1} - 1$ [M2011, eq. (25)]. In [AHHH+2024a] the term $h^{1,1}$ of this formula was read as a Sasakian Hodge number not accessible through the Milnor algebra, and mirror symmetry was suggested as a way to obtain it. It is the basic Hodge number, however, and $h_B^{1,1} = 1$ for every link with $n \geq 3$ by Lemma 3.10, so that $P_B^{1,1} = 0$ and

$$\dim \mathfrak{M}_{CY}(K_f) = 2 + 2h_S^{n-1,1} \qquad (n \geq 3),$$

which is $b_3(K_f)$ for 7-dimensional links; for K3 links, [M2011, eq. (22)] gives

$$\dim \mathfrak{M}_{CY}(K_f) = 3b_2 - 2 = 3h_S^{1,1} + 4.$$

These dimensions are therefore known for every link in the datasets considered here. For $n \geq 3$ the transverse Kähler forms have no moduli, and every deformation comes from the transverse complex structure, whose first-order deformations are the polynomial deformations of X_f (Remark 3.7). This is the moduli-theoretic form of the asymmetry described in Remark 4.7: the classes counted by $h_{CY}^{1,1}$ other than $[d\eta]$ lie in the twisted sectors, which the link does not see.

3.4 The six remaining weight systems

In [AHHH+2024a] the invariants of 6 of the 7555 weight systems were left uncomputed, since the Gröbner-basis computations for them repeatedly exceeded the available walltime, and machine-learning predictions were given in their place. By Corollaries A and 3.6 these invariants are determined by the weights, and they are obtained from (2.6) in a fraction of a second. They are listed in Table 3.1, alongside the predictions of [AHHH+2024a], and they complete the dataset of Sasakian Hodge numbers of 7-dimensional Calabi–Yau links. The distribution of $h_S^{2,1}$ over all 7555 weight systems is shown in Figure 3.1. It takes 287 distinct values between 1 and 491, with median 33; the maximum is attained by $(1, 1, 12, 28, 42)$, one of the six systems of Table 3.1, and exceeds the largest value 416 among the 7549 systems of [AHHH+2024a].

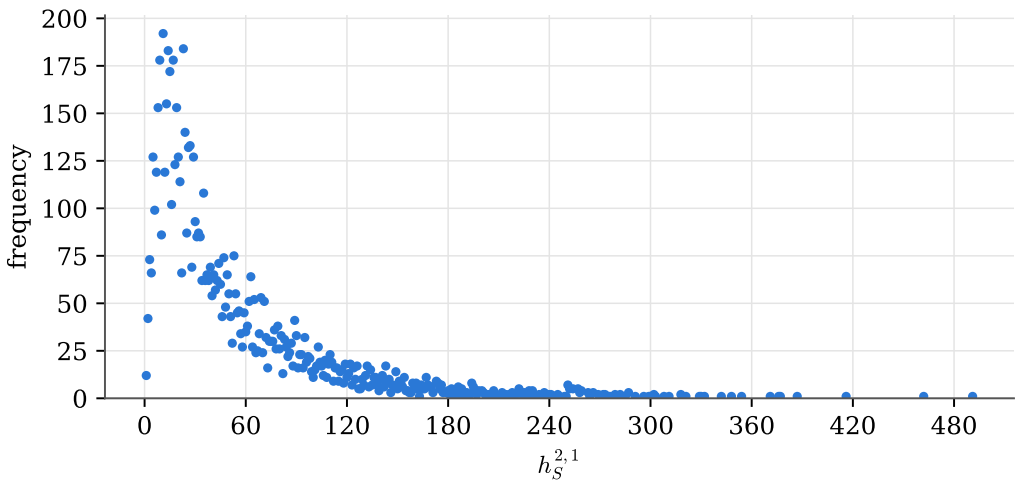


Figure 3.1: Distribution of the Sasakian Hodge number $h_S^{2,1}$ of the 7-dimensional links of the 7555 weighted Calabi–Yau threefolds: each point gives a value of $h_S^{2,1}$ and the number of weight systems attaining it. The remaining Sasakian Hodge numbers are $h_S^{0,0} = h_S^{3,0} = 1$ and zero (Remark 3.14).

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$\mathbf{w}$	$h_S^{2,1}$	$h_{CY}^{2,1}$	$\mu(\mathbf{w})$	ν	NN prediction	SR prediction
$(1,1,8,19,28)^\dagger$	348	348	39788	27	257	338
$(1,1,9,21,32)$	387	387	49665	31	272	377
$(1,1,11,26,39)^\dagger$	462	462	72226	35	293	447
$(1,1,12,28,42)^\dagger$	491	491	82668	37	299	476
$(1,6,34,81,122)^\dagger$	245	246	119805	19	243	243
$(1,6,40,93,140)$	274	275	153714	19	265	272

Table 3.1: The invariants of the six 7-dimensional Calabi–Yau links left open in [AHHH+2024a], computed from Corollary 3.6 and (2.12); in all cases $h_S^{3,0}=1$. The last two columns are the neural-network and symbolic-regression predictions of $h_S^{2,1}$ from [AHHH+2024a]. The rows marked † have since been confirmed by Gröbner bases [AHHH+2024b]; every row has been confirmed by the exact certificates described in the text. The two cases with $h_S^{2,1}<h_{CY}^{2,1}$ each contain a curve of genus one whose generic points are A_1 singularities, in accordance with Theorem D.

Admissible representatives. The exact checks require explicit polynomials whose critical point is isolated. The following elementary criterion supplies the representatives used here and in Section 6.

Lemma 3.16 (Fermat, chain and loop polynomials)**.** *Consider polynomials in N complex variables $z_1,\dots,z_N$. Let j be a map from $\{1,\dots,N\}$ to itself such that every k has at most one preimage other than k itself, and let $a_1,\dots,a_N\geq 2$. Then*

$$f=\sum_{i:\,j(i)=i} z_i^{a_i}+\sum_{i:\,j(i)\neq i} z_i^{a_i}z_{j(i)}$$

has an isolated critical point at the origin. If f is $\mathbf{w}$-homogeneous of degree $d=\sum_i w_i$ with $N\geq 3$, the condition $a_i\geq 2$ holds automatically.

Proof. Every component of the graph of j contains one cycle. If the cycle is a fixed point k, the condition on preimages makes the component a path $i_1\to\cdots\to i_m=k$; otherwise every vertex of the cycle already has a preimage on the cycle, and the component is the cycle. Hence f is a sum, in disjoint sets of variables, of terms of the following forms:

$$\begin{array}{ll}\text{Fermat:} & x^a,\\ \text{chain:} & x_1^{a_1}x_2+\cdots+x_{m-1}^{a_{m-1}}x_m+x_m^{a_m},\\ \text{loop:} & x_1^{a_1}x_2+\cdots+x_{m-1}^{a_{m-1}}x_m+x_m^{a_m}x_1.\end{array}$$

It suffices to treat each summand. For a chain put $M_i=x_i^{a_i}x_{i+1}$ for $i<m$ and $M_m=x_m^{a_m}$; then

$$x_1\partial_1 f=a_1M_1,\qquad x_i\partial_i f=M_{i-1}+a_iM_i\quad(i\geq 2).$$

At a critical point all M_i vanish, so $x_m=0$; then $\partial_m f=x_{m-1}^{a_{m-1}}$, since $a_m\geq 2$, gives $x_{m-1}=0$, and in the same way $\partial_i f=x_{i-1}^{a_{i-1}}+a_ix_i^{a_i-1}x_{i+1}$ gives $x_{i-1}=0$ for $i=m-1,\dots,2$. For a loop put $M_i=x_i^{a_i}x_{i+1}$ with indices modulo m; then $x_i\partial_i f=M_{i-1}+a_iM_i$, so at a critical point

$$M_i=(-1)^m\Big(\prod_j a_j\Big)M_i,$$

which forces $M_i=0$ as $\prod_j a_j>1$. Hence some x_i vanishes, and $\partial_i f=x_{i-1}^{a_{i-1}}+a_ix_i^{a_i-1}x_{i+1}$ gives $x_{i-1}=0$, and so on around the loop. Finally, if $d=\sum_i w_i$ and $N\geq 3$, then $a_iw_i=d-w_{j(i)}>w_i$ for $j(i)\neq i$, and $a_i=d/w_i>1$ for $j(i)=i$. □

These are the invertible polynomials of Kreuzer and Skarke [KS1992a]. The invertible polynomials used here, such as the one displayed below, were found by a randomised search for a map j compatible with $(\mathbf{w},d)$, and they have all coefficients equal to 1.

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Verification and comparison. The values have been confirmed in two further ways. Firstly, Gröbner-basis results for four of the six systems were added to the public dataset of [AHHH+2024b] in December 2024, and they agree with Table 3.1 in both $h_S^{2,1}$ and ν. That dataset contains no values for the remaining two systems, $(1,1,9,21,32)$ and $(1,6,40,93,140)$, for which the values in Table 3.1 are new. Secondly, for each of the six systems the Hilbert function of $\mathbb{M}_f$ was certified exactly in every degree $0 \le \ell \le 3d$ for an explicit invertible polynomial, such as

$$f = z_1^{280} + z_1 z_4^3 + z_2^{40} z_3 + z_3^7 + z_5^2 \qquad \text{for } \mathbf{w} = (1,6,40,93,140),$$

which is admissible by Lemma 3.16. The certificate, described in Appendix A.1, is exact linear algebra over $\mathbb{Q}$ on the degree-ℓ part of the Koszul complex of $\partial_1 f, \dots, \partial_5 f$, checked by ranks modulo a prime; it uses neither a Gröbner basis nor Theorem 3.2. The certified Hilbert functions agree with the coefficients of $P_{\mathbf{w}}$, their sums equal $\mu(\mathbf{w})$, and through (2.12) they give the values of ν in Table 3.1. In addition, $h_S^{2,1} = \dim(\mathbb{M}_g)_d$ was certified in the same way for a second polynomial g in each case, with all monomials of degree d and coefficients drawn uniformly from $\{1,\dots,999\}$, after the singularity of g had been certified to be isolated (Appendix A.1). These computations take about 80 seconds in total.

Two features of these systems help explain the earlier predictions. They are not those of largest Milnor number, as $\mu(\mathbf{w})$ reaches 909720 among the systems computed in [AHHH+2024a]; however, they include the two largest values of $h_S^{2,1}$ among all 7555 weight systems, 491 for $(1,1,12,28,42)$ and 462 for $(1,1,11,26,39)$, whereas the largest value in the dataset of [AHHH+2024a] is 416. The neural network underestimates all six values, by as much as 192, and two of them lie outside the range of the data on which it was trained. The symbolic-regression formula, which was observed in [AHHH+2024a] to predict values very close to the bound $h_{CY}^{2,1}$, is within 15 of the exact value in every case. This behaviour is explained by the exact values: the bound $h_S^{2,1} \le h_{CY}^{2,1}$ of the Calabi–Yau Bound Conjecture, proved in Section 4, is attained in four of the cases and missed by one in the other two, where the defect comes from the genus-one curves $X_f \cap \mathbb{P}(6,34,122)$ and $X_f \cap \mathbb{P}(6,40,140)$, whose generic points have stabiliser μ_2.

4 Proof of the Calabi–Yau Bound Conjecture

Throughout this section $(\mathbf{w}, d)$ is Calabi–Yau, $d = \sum_i w_i$, $f \in U_{\mathbf{w}}$, $\mathcal{X} = \mathcal{X}_f$ and $X = X_f$. Building on Lemmas 3.8 and 3.10, we compare the three Hodge theories appearing in (1.4). The idea of the proof is that the Calabi–Yau Hodge numbers of the Kreuzer–Skarke list are *stringy* Hodge numbers of X, which decompose into the Hodge numbers of the orbifold $\mathcal{X}$ and contributions from its loci of orbifold points; the former are the Sasakian Hodge numbers, up to the diagonal correction of (1.4).

4.1 Orbifold and stringy Hodge numbers

For $t \in \mathbb{C}^*$ of finite order put

$$S(t) = \{i : t^{w_i} = 1\}, \qquad Y^t = \{z \in Y : t \cdot z = z\} = Y \cap \mathbb{C}^{S(t)}, \tag{4.1}$$

and write $\mathbf{w}_S = (w_i)_{i \in S}$ for $S \subseteq \{1,\dots,N\}$. Since $\mathbb{C}^*$ is abelian, the *sector* $\mathcal{X}_t = [Y^t/\mathbb{C}^*]$ is a closed suborbifold of $\mathcal{X}$, and the inertia orbifold of $\mathcal{X}$ is the disjoint union of the sectors $\mathcal{X}_t$ over the t of finite order with $Y^t \neq \emptyset$. The coarse space of $\mathcal{X}_t$ is the locus $X^t = X \cap \mathbb{P}(\mathbf{w}_{S(t)})$, but $\mathcal{X}_t$ keeps its stabilisers. In particular, if $e = \gcd(\mathbf{w}_{S(t)}) > 1$, then μ_e acts trivially on Y^t, and $\mathcal{X}_t$ is a μ_e-gerbe over the quotient of Y^t by $\mathbb{C}^*/\mu_e \cong \mathbb{C}^*$, which acts with the weights $\mathbf{w}_{S(t)}/e$.

The contribution of a sector is shifted by the age of the stabiliser action. The following tangent-space calculation fixes this shift directly from the weights.

Lemma 4.1 (ages)**.** *Let $f \in U_{\mathbf{w}}$ have Calabi–Yau degree $d = \sum_i w_i$, and write $Y = f^{-1}(0) \setminus 0$ and $\mathcal{X} = [Y/\mathbb{C}^*]$. Let $z \in Y$ and let $t = e^{2\pi i\theta}$ belong to the stabiliser of z. The eigenvalues of t on the tangent space of $\mathcal{X}$ at z are the multiset $\{t^{w_i}\}_{i=1}^N$ with one eigenvalue t^d and one eigenvalue 1 removed. Hence the age of t, the sum of the numbers $\vartheta \in [0,1)$ such that $e^{2\pi i\vartheta}$ is an eigenvalue, is*

$$\operatorname{age}(t) = \sum_{i=1}^{N} \langle \theta w_i \rangle - \langle \theta d \rangle, \qquad \langle x \rangle = x - \lfloor x \rfloor,$$

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which depends only on t, and equals $\sum_{i\notin S(t)}\langle\theta w_i\rangle$ if $t^d=1$.

Proof. The tangent space of $\mathcal{X}$ at z is $\ker(df_z)/\mathbb{C}E_{\mathbf{w}}(z)$, with $E_{\mathbf{w}}(z)=(w_iz_i)_i$. Differentiating $f(t\cdot z)=t^df(z)$ at the fixed point z gives

$$df_z\circ t=t^d\,df_z,$$

so passing to $\ker df_z$ removes one eigenvalue t^d from those of t on $\mathbb{C}^N$, and quotienting by the fixed line $\mathbb{C}E_{\mathbf{w}}(z)$ removes one eigenvalue 1. □

For $F\subseteq\{1,\dots,N\}$ write $g_F=\gcd(w_i:i\in F)$. Let $J=e^{2\pi i/d}\in\mathbb{C}^*$, and for $k\in\mathbb{Z}/d$ put $F_k=S(J^k)=\{i:d\mid kw_i\}$.

We next determine which fixed loci can occur. Part (iv) will count the sectors supported on each codimension-two locus in the defect formula.

Lemma 4.2 (the twisted sectors)**.** *Let $f\in U_{\mathbf{w}}$ have degree $d=\sum_i w_i$ in $N=n+2$ variables, and write $Y=f^{-1}(0)\setminus 0$ and $X=X_f$. For a finite-order element $t\in\mathbb{C}^*$, put $S(t)=\{i:t^{w_i}=1\}$, $Y^t=Y\cap\mathbb{C}^{S(t)}$ and $X^t=X\cap\mathbb{P}(\mathbf{w}_{S(t)})$. Assume $t\neq1$ and $Y^t\neq\emptyset$. As above, set $J=e^{2\pi i/d}$, $F_k=S(J^k)$ and $g_F=\gcd(w_i:i\in F)$.*

(i) If $t^d\neq1$, then f vanishes on $\mathbb{C}^{S(t)}$, $|S(t)|\le N/2$, and $X^t=\mathbb{P}(\mathbf{w}_{S(t)})$ is a weighted projective space.

(ii) If $t^d=1$, then $f|_{\mathbb{C}^{S(t)}}$ has an isolated critical point at the origin, so X^t is a quasi-smooth hypersurface of dimension $|S(t)|-2$ in $\mathbb{P}(\mathbf{w}_{S(t)})$.

(iii) If $t=J^k$, then $|F_k|\neq N-1$, and $\operatorname{age}(t)=1$ whenever $|F_k|=n$.

(iv) If $n\ge3$, $|F|=n$ and $g_F>1$, then $g_F\mid d$, and exactly g_F-1 classes $k\in\mathbb{Z}/d$ satisfy $F_k=F$.

Proof. (i) Every monomial in the variables indexed by $S(t)$ is fixed by t; one of degree d would force $t^d=1$, so $f|_{\mathbb{C}^{S(t)}}\equiv0$. Then $\mathbb{C}^{S(t)}\setminus0\subset Y$, and smoothness of Y requires the $N-|S(t)|$ polynomials $\partial_jf|_{\mathbb{C}^{S(t)}}$, $j\notin S(t)$, to have no common zero on $\mathbb{C}^{S(t)}\setminus0$; as they are homogeneous of positive degree, this forces $N-|S(t)|\ge|S(t)|$. (ii) If $t^d=1$ and $z\in Y^t$, then $df_z\circ t=df_z$, so df_z lies in the span of the dz_i with $i\in S(t)$, and the differential of $f|_{\mathbb{C}^{S(t)}}$ at z is nonzero. By Euler's identity (2.4) this excludes critical points other than the origin; compare [FJR2013, Lemma 2.1.10]. (iii) By Lemma 4.1 the age is $\sum_{i\notin F_k}\langle kw_i/d\rangle$, an integer since $\sum_i kw_i/d=k$. If $|F_k|=N-1$ the single remaining fractional part would be a nonzero integer. If $|F_k|=n$ there are two fractional parts in $(0,1)$ with integer sum, so the age is 1. (iv) If $g_F\nmid d$, no monomial of degree d involves only the variables in F, so $f|_{\mathbb{C}^F}\equiv0$ and the argument of (i) gives $n\le N/2$, that is $n\le2$. For $n\ge3$ therefore $g_F\mid d$. Now $F_k\supseteq F$ if and only if $k\in\frac{d}{g_F}\mathbb{Z}$, which gives g_F classes; for $k\neq0$ among them $|F_k|\in\{n,n+1,n+2\}$, the value $n+1$ is excluded by (iii), and $n+2$ forces $k=0$ since $\gcd(\mathbf{w})=1$. □

Lemmas 4.1 and 4.2 determine the local structure of $\mathcal{X}$. Every $t\neq1$ in the stabiliser of a point $z\in Y$ acts non-trivially on the tangent space of $\mathcal{X}$ at z. Otherwise, by Lemma 4.1, either $t^d\neq1$ and $|S(t)|=N-1$, which contradicts Lemma 4.2(i), or $t^d=1$ and $|S(t)|=N$, so that $t=1$ as $\gcd(\mathbf{w})=1$. The determinant of t on the tangent space is $t^{\sum_i w_i-d}=1$, so no t acts as a reflection, which would have exactly one eigenvalue different from 1, equal to its determinant. Hence $\mathcal{X}$ is an effective orbifold without reflections, its coarse space X has Gorenstein canonical cyclic quotient singularities, and every $t\neq1$ with $Y^t\neq\emptyset$ has positive integer age. Following Chen and Ruan [CR2004, Def. 3.2.4], the *orbifold Hodge numbers* of $\mathcal{X}$ are

$$h^{p,q}_{\mathrm{orb}}(\mathcal{X})=\sum_t h^{\,p-\operatorname{age}(t),\,q-\operatorname{age}(t)}(\mathcal{X}_t),\tag{4.2}$$

where t runs over the elements of finite order with $Y^t\neq\emptyset$, and $h^{i,j}(\mathcal{X}_t)=\dim H^j(\mathcal{X}_t,\Omega^i_{\mathcal{X}_t})$ are the ordinary Hodge numbers of the sector, which by Section 2.2 are those of its coarse space X^t. A sector that is disconnected contributes the sum over its components, on which the age is the same by Lemma 4.1. The term $t=1$ is the *untwisted sector* $\mathcal{X}_1=\mathcal{X}$, and the others are the *twisted sectors*. The gerbe structure of a sector does not change its Hodge numbers, since the generic stabiliser acts trivially on its differential forms; how often a given locus occurs is recorded only by the number of elements t with that fixed locus. Since $\mathcal{X}$ has no reflections and X has Gorenstein quotient singularities, Yasuda's theorem [Y2004, Thm. 3.15 and the

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proof of Cor. 3.16] identifies (4.2) with Batyrev's stringy Hodge numbers $h^{p,q}_{st}(X)$ [B1998, Def. 3.8] (see also [Y2004, Rem. 1.4]), and these are the ordinary Hodge numbers of any crepant resolution when one exists [B1998, Thm. 3.12]. For $n = 3$ the Kreuzer–Skarke values were computed with Vafa's Landau–Ginzburg orbifold formula [V1989, KS1994, KS1992b], and for $n = 4$ in [LSW1999] by the same method. By [B2020, Prop. 4.3, Cor. 4.4] the generating function of that formula is

$$(-u)^n E_{\mathrm{orb}}(\mathcal{X}; u^{-1}, v) = \sum_{p,q} (-1)^{p+q} h^{n-p,q}_{st}(X)\, u^p v^q,$$

in which the Hodge diamond is rotated; the tabulated $h^{p,1}_{CY}$ are read off from the coefficients of $u^{n-p}v$, and they are the stringy Hodge numbers $h^{p,1}_{st}(X)$. For $n = 3$ the same values are the Hodge numbers of Batyrev's toric crepant resolution [B1994, B1998]. In what follows $h^{p,q}_{CY} = h^{p,q}_{st}(X)$.

The untwisted term can now be read in the Sasakian notation of Definition 2.1.

Lemma 4.3 (the untwisted sector)**.** *Let K_f be the link of an admissible Calabi–Yau polynomial in $N = n+2$ variables, and let $\mathcal{X} = \mathcal{X}_f$ be its quotient orbifold. For $p + q = n$, the untwisted contribution to (4.2) is*

$$h^{p,q}(\mathcal{X}) = h^{p,q}_S(K_f) + \delta_{pq}. \tag{4.3}$$

Proof. By Lemma 3.8, $h^{p,q}(\mathcal{X}) = h^{p,q}_B(K_f)$, and by Definition 2.1 and Lemma 3.10, $h^{p,q}_B = h^{p,q}_S + \delta_{pq}$ for $p + q = n$. □

4.2 The bound and its defect

Lemma 4.3 identifies the part of the stringy Hodge numbers detected by the link. The other sectors give a non-negative correction, and Lemma 4.2 identifies exactly those that can contribute to $h^{n-1,1}$.

Proof of Theorem C. (i) In (4.2), the untwisted term is $h^{p,q}_S + \delta_{pq}$ by Lemma 4.3, and every twisted term is a Hodge number of a compact orbifold and hence non-negative.

(ii) A twisted sector with $\mathrm{age}(t) = c \geq 1$ contributes $h^{n-1-c,1-c}(\mathcal{X}_t)$ to the $(n-1,1)$ entry, which vanishes unless $c = 1$; so only the numbers $h^{n-2,0}(\mathcal{X}_t)$ of sectors of age one and dimension at least $n-2$ contribute. If $t^d \neq 1$, then X^t is a weighted projective space by Lemma 4.2(i), for which $h^{n-2,0} = 0$ as $n \geq 3$. If $t^d = 1$, then $t = J^k$ for some k, and $\dim \mathcal{X}_t = |F_k| - 2 \geq n-2$ forces $|F_k| = n$, since $|F_k| = n+1$ is excluded by Lemma 4.2(iii) and $|F_k| = N$ would force $t = 1$ as $\gcd(\mathbf{w}) = 1$; the age is then 1. Fix such an $F = F_k$, and put $r = n-2 \geq 1$ and $g = g_F$, so that $\mathcal{X}_t = \mathcal{C}_F$. By Lemma 4.2(ii) and (iv), $\mathcal{C}_F$ is a μ_g-gerbe over the quotient of the quasi-smooth hypersurface $\{f|_{\mathbb{C}^F} = 0\} \setminus 0$ by $\mathbb{C}^*$ acting with the weights $\mathbf{w}_F/g$, which have greatest common divisor 1, and $f|_{\mathbb{C}^F}$ has degree d/g for these weights. Neither the gerbe nor the rescaling changes Hodge numbers, and Lemmas 3.11 and 3.12 apply to the rescaled hypersurface with r in place of n. Its holomorphic r-forms are therefore the sections of $\mathcal{O}(\kappa_F)$ with $\kappa_F = (d - \sum_{i\in F} w_i)/g$, and since $\kappa_F < d/g$ there is no correction from the equation:

$$h^{r,0}(\mathcal{C}_F) = p_{\mathbf{w}_F/g}(\kappa_F) = p_{\mathbf{w}_F}\Big(d - \sum_{i\in F} w_i\Big).$$

Summing over the $g_F - 1$ elements $t = J^k$ with $F_k = F$ of Lemma 4.2(iv) gives (1.5). Finally, a point $z \in Y$ has stabiliser containing μ_{g_F} exactly when $z_i = 0$ for every i with $g_F \nmid w_i$, and the set of such i is the complement of F, since by Lemma 4.2(iii) and $\gcd(\mathbf{w}) = 1$ no further weight is divisible by g_F. □

For $n = 3$ the sectors $\mathcal{C}_F$ with $|F| = 3$ are curves, and the summand of (1.5) is $g_F - 1$ times their genus. This proves the Calabi–Yau Bound Conjecture and describes exactly when it is sharp.

Proof of Theorem D. The identity is Theorem C(ii) with $n = 3$, using $h^{2,1}_{CY} = h^{2,1}_{st}(X_f)$. The coarse space of a one-dimensional orbifold is a smooth curve, C_F is connected because $h^0(\mathcal{O}) = 1$ by Lemma 3.11, and its genus $h^{1,0}$ is $p_g(C_F)$ by the proof of Theorem C; hence $p_g(C_F) = 0$ exactly when $C_F \cong \mathbb{P}^1$. The polynomial $f|_{\mathbb{C}^F}$ is not a power of a single variable, since that would not have an isolated critical point, so the points of C_F with all coordinates in F nonzero form a dense open subset, and their stabiliser is μ_{g_F}. At such a

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point, write $\{1,\dots,5\}\setminus F=\{j,j'\}$ and let t generate μ_{g_F}. By Lemma 4.1 the eigenvalues of t on the tangent space are 1, along the curve, and t^{w_j}, $t^{w_{j'}}$, whose product is $t^{d-\sum_{i\in F}w_i}=1$. A prime dividing g_F and w_j would also divide $w_{j'}$, and hence every weight, so t^{w_j} is a primitive g_F-th root of unity and the transversal singularity is of type A_{g_F-1}. □

The identity was evaluated for all 7555 weight systems (Appendix A): equality holds for exactly 4202 of them, and for 4198 of the 7549 considered in [AHHH+2024a].

Example 4.4. For $\mathbf{w}=(1,6,34,81,122)$, of degree 244, the only index set F with $|F|=3$ and $g_F>1$ is the one of the weights $6,34,122$, with $g_F=2$. The curve $C_F\subset\mathbb{P}(6,34,122)$ has genus $p_{(6,34,122)}(82)=1$, since $82=8\cdot 6+34$ is the only representation, and therefore $h^{2,1}_{CY}-h^{2,1}_S=1$, in agreement with Table 3.1. The curve passes through the coordinate point of weight 34, which lies on X_f for every f because $34\nmid 244$, and whose stabiliser is μ_{34}; the genus in Theorem D is that of the whole curve.

Remark 4.5. The identity of Theorem D was also verified numerically for all 7555 weight systems, using the Kreuzer–Skarke values of $h^{2,1}_{CY}$ [KS2026a], the Gröbner-basis values of $h^{2,1}_S$ from [AHHH+2024a] for the 7549 systems computed there, and Table 3.1 for the other six. The defect is related to the classical distinction between polynomial and non-polynomial deformations [GH1987, COK1995]: $h^{2,1}_S$ counts the polynomial deformations of X_f (Remark 3.7), so the right-hand side of (1.5) is the number of the remaining ones. For instance, for $\mathbb{P}(1,1,2,2,2)[8]$ the value $h^{2,1}_{CY}=86$ consists of 83 polynomial deformations and 3 attached to the curve of genus 3 whose generic points are A_1 singularities. Candelas, de la Ossa and Katz [COK1995, §5.2] also identify 15 non-polynomial deformations of $\mathbb{P}(1,2,2,2,7)[14]$ with the genus of its singular curve. Theorem D recovers that contribution as $(2-1)p_{(2,2,2)}(8)=15$. More generally, a correction from codimension-two strata already appears in Batyrev's toric formula [B1994, Thm. 4.3.7], under its reflexive-polytope hypotheses. Theorem C gives the Sasakian interpretation for admissible weighted hypersurfaces: it identifies the polynomial contribution with $h^{n-1,1}_S$, expresses the correction directly in terms of weights and stabilisers, and applies without choosing a crepant resolution.

In [HS2024, §5.1] an approximation to Vafa's formula, in the form of [B2020], was studied that keeps the term of age zero and the terms of maximal size; it gives the exact value of $h^{2,1}$ for 56% of the 7555 threefolds [HS2024, Table 5.1]. The term of age zero equals $\sum_k a_{kd}(uv)^k$, so in the rotated grading above it contributes $a_d=h^{n-1,1}_S$ to $h^{n-1,1}$, while the terms of maximal size contribute only to diagonal entries. For $n\geq 3$ that approximation to $h^{n-1,1}$ is therefore exactly $h^{n-1,1}_S$, and the omitted terms that change it are those of the sectors $\mathcal{C}_F$ of Theorem C. Consequently, Theorem C recovers the $h^{1,2}$ and $h^{1,3}$ lower bounds in [HS2024, eqs. (5.6)–(5.7)], using Hodge symmetry, and gives their exact error and equality criterion; it also gives the corresponding bound for every $n\geq 3$. The observation of [HS2024] is thus the equality case of Theorem D, with proportion $4202/7555\approx 55.6\%$, and for fourfolds the proportion 48.6% of Section 6 matches the 48.7% found in [HS2024, Table 5.2] for a sample of about 220 000 weight systems.

For fourfolds Theorem C gives the analogous statement, with surfaces in place of curves.

Corollary 4.6 ($n=4$)**.** *Let K_f be the nine-dimensional link of an admissible Calabi–Yau polynomial in six variables, and let X_f be the corresponding fourfold. With the stringy Hodge numbers and fixed-locus notation of Theorem C,*

$$h^{3,1}_{CY}-h^{3,1}_S=\sum_{|F|=4,\ g_F>1}(g_F-1)\,p_g(\mathcal{C}_F)\qquad\text{and}\qquad h^{2,2}_S\leq h^{2,2}_{CY}-1,$$

where the surfaces $\mathcal{C}_F$ have coarse spaces $X_f\cap\mathbb{P}(\mathbf{w}_F)$ and $p_g(\mathcal{C}_F)=p_{\mathbf{w}_F}(d-\sum_{i\in F}w_i)$. Here

$$h^{2,2}_{CY}=44+4h^{1,1}_{CY}-2h^{2,1}_{CY}+4h^{3,1}_{CY}. \tag{4.4}$$

Proof. The first two statements are Theorem C(ii) and (i). For the last, X_f is projective with Gorenstein canonical singularities and trivial canonical class, and its stringy Hodge numbers exist, so the stringy Libgober–Wood identity holds [B2000, Cor. 3.10, Cor. 3.11]:

$$\sum_{p,q}(-1)^{p+q}(p-2)^2h^{p,q}_{st}=\frac{1}{3}e_{st},\qquad e_{st}=\sum_{p,q}(-1)^{p+q}h^{p,q}_{st}. \tag{4.5}$$

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Twisted sectors have positive age and therefore do not contribute to $h^{p,0}_{st}$, so $h^{0,0}_{st} = h^{4,0}_{st} = 1$ and $h^{p,0}_{st} = 0$ for $0 < p < 4$ by Lemma 3.10 and Corollary 3.6. With the symmetries $h^{p,q}_{st} = h^{q,p}_{st} = h^{4-p,4-q}_{st}$, the identity becomes the stated formula, exactly as for smooth Calabi–Yau fourfolds. □

Restricted to the untwisted sector, the same identity relates the two Sasakian Hodge numbers of a 9-dimensional link, up to a correction from the twisted sectors.

Proof of Theorem E. Write the decomposition (4.2) into the untwisted sector and the sum of the twisted ones as

$$\begin{aligned} h^{p,q}_{CY} &= h^{p,q}_{\mathrm{untw}} + T^{p,q}, \\ T^{p,q} &= \sum_{t\neq 1} h^{p-\mathrm{age}(t),q-\mathrm{age}(t)}(\mathcal{X}_t). \end{aligned} \tag{4.6}$$

By Lemmas 4.3 and 3.10 the untwisted parts of $h^{1,1}$, $h^{2,1}$, $h^{3,1}$ and $h^{2,2}$ are 1, 0, $h^{3,1}_S$ and $h^{2,2}_S + 1$. Inserting this into (4.4) gives (1.6) with

$$R = 4T^{1,1} - 2T^{2,1} + 4T^{3,1} - T^{2,2},$$

so that a sector of age a contributes

$$\rho(t) = 4h^{1-a,1-a} - 2h^{2-a,1-a} + 4h^{3-a,1-a} - h^{2-a,2-a}$$

of its own Hodge numbers. By Lemma 4.2(i) and (iii), $Y^t = Y \cap \mathbb{C}^{S(t)}$ with $|S(t)| \le 3$ if $t^d \neq 1$ and $|S(t)| \le 4$ if $t^d = 1$, so $m \le 2$. The eigenvalues of t^{-1} on the normal space of $\mathcal{X}_t$ are the inverses of those of t, none equal to 1, so

$$\mathrm{age}(t) + \mathrm{age}(t^{-1}) = 4 - m, \tag{4.7}$$

and both ages are at least 1. Hence $a = 1$ if $m = 2$, $\{\mathrm{age}(t), \mathrm{age}(t^{-1})\} = \{1, 2\}$ if $m = 1$, and $a \in \{1, 2, 3\}$ if $m = 0$. For $m = 2$, Hodge symmetry and Poincaré duality on the coarse space give

$$\begin{aligned} \rho &= 4h^{0,0} - 2h^{1,0} + 4h^{2,0} - h^{1,1} \\ &= 6(h^{0,0} - h^{0,1} + h^{0,2}) - \sum_{p,q} (-1)^{p+q} h^{p,q} \\ &= 6\chi(\mathcal{O}) - e. \end{aligned}$$

For $m = 1$ the sector is connected (Lemmas 4.2 and 3.11), so $\rho = 4 - 2g - 1 = e + 1$ for age one, where $g = h^{1,0}$, and $\rho = -h^{0,0} = -1$ for age two. For $m = 0$, $\rho = 4h^{0,0}$, $-h^{0,0}$ or 0 for ages 1, 2 or 3. If X_f has no orbifold points there are no twisted sectors and $R = 0$. □

The identity has been checked independently of the Kreuzer–Skarke data for 2068 weight systems: 2000 drawn at random and the 25 largest values of $h^{3,1}_S$, of R and of $-R$. For each of them every twisted sector was constructed from the weights by Lemmas 4.1 and 4.2, and its Hodge numbers were computed by Lemma 3.11 and Theorem B for surfaces and curves, and by counting points for sectors of dimension zero. In every case the sum of the $\rho(t)$ equals $h^{2,2}_S - 4h^{3,1}_S - 47$ computed from (6.1), and the resulting twisted parts reproduce the Kreuzer–Skarke values of $h^{1,1}_{CY}$, $h^{2,1}_{CY}$ and $h^{3,1}_{CY}$ (script `lw_identity_9d.py` of the GitHub repository).

Remark 4.7 (mirror symmetry)**.** The Sasakian Hodge numbers of Calabi–Yau links do not exhibit mirror symmetry, and Theorem C shows why. For Calabi–Yau n-folds mirror symmetry exchanges $h^{p,q}$ with $h^{n-p,q}$, and in particular $h^{n-1,1}$ with $h^{1,1}$. For $n \ge 3$, however, $h^{1,1}_S = 0$ for every link by Remark 3.14, whereas $h^{n-1,1}_S \ge 1$ for each of the 7555 links with $n = 3$ and the 1 100 055 links with $n = 4$ considered here; hence the Sasakian Hodge numbers of no link in these datasets are the mirror image of those of another link, or of its own. The underlying reason is that the link sees only the untwisted sector of (4.2) (Lemma 4.3), and mirror symmetry does not preserve the splitting into untwisted and twisted sectors. To $h^{n-1,1}$ the untwisted sector contributes exactly the polynomial deformations, counted by $h^{n-1,1}_S$ (Remark 3.7), while by Lemma 3.10 its contribution to $h^{1,1}$ is only the class $[d\eta]$, which is proportional to the hyperplane class; every further class of type $(1,1)$ comes from a twisted sector of age one, since a sector of age c contributes $h^{1-c,1-c}$ to $h^{1,1}$ in (4.2). Suppose, for instance, that the mirror of X_f is itself a quasi-smooth Calabi–Yau hypersurface X_g in a weighted projective space, so that $h^{p,q}_{CY}(X_g) = h^{n-p,q}_{CY}(X_f)$. By (4.2) and Lemma 3.10, $h^{1,1}_{CY}(X_g) = 1 + \tau(\mathcal{X}_g)$,

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where $\tau(\mathcal{X}_g)$ is the number of connected components of the twisted sectors of age one of $\mathcal{X}_g$, and Theorem C becomes

$$h_S^{n-1,1}(K_f) + \sum_{|F|=n,\ g_F>1} (g_F - 1)\, p_{\mathbf{w}_F}\Big(d - \sum_{i\in F} w_i\Big) = 1 + \tau(\mathcal{X}_g).$$

The left-hand side is computed on the link of X_f, while the right-hand side, apart from the hyperplane class, is invisible to the link of X_g, whose own $h_S^{1,1}$ vanishes. Mirror symmetry is therefore a statement about the stringy Hodge numbers $h_{CY}^{p,q}$, not about the Sasakian Hodge numbers of the links. At that level every quasi-smooth Calabi–Yau hypersurface X_f has a mirror $X^\vee$ with $h_{st}^{p,q}(X^\vee) = h_{CY}^{n-p,q}(X_f)$ [B1994, B2020], but $X^\vee$ is a compactified toric hypersurface that is in general not a hypersurface in a weighted projective space, so the lists of weight systems are not closed under the exchange [CLS1990, KS1992b], and in general the mirror has no link of the kind considered here.

5 Five-dimensional links

We apply Corollary 3.6 and Theorem C to links whose quotient orbifold has complex dimension $n = 2$. The Hodge numbers determine the second Betti number, while the twisted sectors record the exceptional curves of the K3 resolution. The complete list is given in Appendix A.2.

For $n = 2$ the 95 Calabi–Yau weight systems of quasi-smooth K3 surfaces [Y1990, IF2000] give 5-dimensional links K_f. They are simply connected Sasakian 5-manifolds whose leaf space is a K3 orbifold. By Corollary 3.6 and Lemma 3.10 their Sasakian Hodge numbers and second Betti number are

$$h_S^{2,0} = h_S^{0,2} = 1, \qquad h_S^{1,1} = p_{\mathbf{w}}(d) - \sum_{i=1}^{4} p_{\mathbf{w}}(w_i), \qquad b_2(K_f) = 2 + h_S^{1,1}. \tag{5.1}$$

Together with $h_S^{0,0} = 1$ and $h_S^{1,0} = h_S^{0,1} = 0$ these are all the Sasakian Hodge numbers of the link (Remark 3.14). The number $h_S^{1,1}$ is related to the singularities of the K3 surface X_f by a simple count. Every singular point x of X_f is a cyclic quotient singularity $\mathbb{C}^2/\mu_{k_x}$ with $\mu_{k_x} \subset \mathrm{SL}_2(\mathbb{C})$, by Lemma 4.1 and the local-structure discussion following Lemma 4.2, that is, a Du Val singularity of type A_{k_x-1}, whose minimal resolution has $k_x - 1$ exceptional curves.

Proposition 5.1. *Let K_f be the five-dimensional link of an admissible Calabi–Yau polynomial in four variables, and let X_f be its K3 coarse space. For each singular point $x \in X_f$, write A_{k_x-1} for its Du Val type. Then*

$$20 = 1 + h_S^{1,1}(K_f) + \sum_{x\in \mathrm{Sing} X_f} (k_x - 1). \tag{5.2}$$

Proof. The minimal resolution of X_f is crepant and is a K3 surface, so $h_{st}^{1,1}(X_f) = 20$ [B1998, Thm. 3.12]. In (4.2) the untwisted sector contributes $h^{1,1}(\mathcal{X}_f) = 1 + h_S^{1,1}$, as in Lemma 4.3. Each $t \neq 1$ in the stabiliser μ_{k_x} of a point x acts on the tangent plane non-trivially and with determinant 1, so without eigenvalue 1; its fixed points are therefore isolated, and it has age 1. The twisted sectors are thus points, one for each pair (x, t) with $t \in \mu_{k_x} \setminus \{1\}$, and each contributes $h^{0,0} = 1$ to $h_{st}^{1,1}$. Summing over all singular points gives $\sum_x (k_x - 1)$. □

The data are collected in Table A.2 in Appendix A.2. In the authors' follow-up computations $(h_S^{2,0}, h_S^{1,1})$ was computed with Gröbner bases for one explicit polynomial for each of the 95 weight systems (file `K3Topological_Data.txt` of the GitHub repository), and these values agree with (5.1) in all 95 cases. Each of these polynomials was certified to have an isolated singularity (Appendix A.1), and by Theorem A the values hold for every admissible polynomial. The singular points were read off from the weights. For generic f and each set S of coordinates with $g_S > 1$, the points of X_f whose nonzero coordinates are exactly those in S number $p_{\mathbf{w}_S}(d) - 1$ when $|S| = 2$, and one or zero when $|S| = 1$ according to whether $w_i \nmid d$ or $w_i \mid d$; the positions of these points depend on f. The identity of Proposition 5.1 holds for all 95 systems. Across the dataset $h_S^{1,1}$ takes every value from 1 to 19 except 15, with the distribution shown in Figure 5.1.

For example the K3 surface $X_{27} \subset \mathbb{P}(5, 6, 7, 9)$ has singularities $A_6 + A_5 + A_4 + A_2$, hence 17 exceptional curves, and its link has $h_S^{1,1} = 2$.

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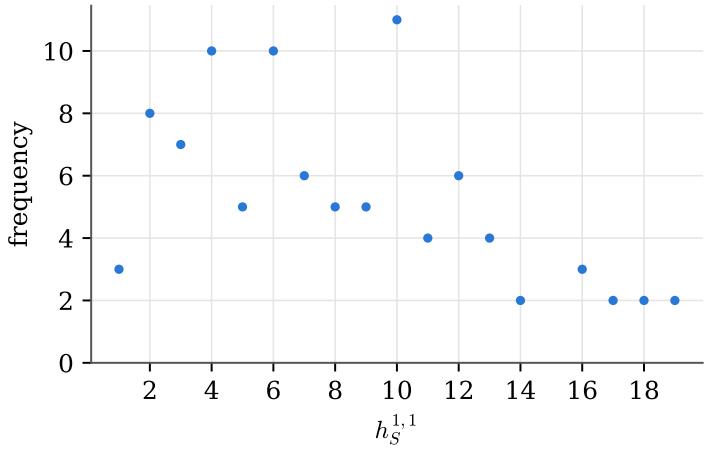


Figure 5.1: Distribution of the Sasakian Hodge number $h_S^{1,1}$ of the 5-dimensional links of the 95 weighted K3 surfaces: each point gives a value of $h_S^{1,1}$ and the number of weight systems attaining it. The remaining Sasakian Hodge numbers are $h_S^{0,0} = h_S^{2,0} = 1$ and $h_S^{1,0} = 0$ (Remark 3.14).

Remark 5.2 (topology of K3 links)**.** The topology of these links is known [BGM2006]. A K3 link is spin, since it is the boundary of the Milnor fibre, which is parallelisable, so that K_f is stably parallelisable. Its second homology is torsion-free [BGM2006, Thm. 39], so by Smale's classification of simply connected spin 5-manifolds [S1962]

$$H_2(K_f;\mathbb{Z}) \cong \mathbb{Z}^{\,2+h_S^{1,1}}, \qquad K_f \cong \#_{\,2+h_S^{1,1}}(S^2 \times S^3),$$

as observed in [BGM2006, Ex. 41]. Table A.2 recovers the range $3 \le b_2 \le 21$, $b_2 \ne 17$, recorded in [BGM2006, Cor. 42]; the absent value $b_2 = 17$ corresponds to the absent value $h_S^{1,1} = 15$.

6 Nine-dimensional links

We now use Corollary 3.6 to compute the middle Sasakian Hodge numbers for the complete fourfold list. Corollary 4.6 provides the comparison with stringy Hodge numbers, and the independent checks below test the calculation on weights and on explicit polynomials.

For $n = 4$ the link K_f is a 3-connected 9-manifold whose leaf space is a Calabi–Yau fourfold orbifold $\mathcal{X}_f$, with coarse space $X_f \subset \mathbb{P}^5(\mathbf{w})$. By Corollary 3.6 and (3.1), its Sasakian Hodge numbers and middle Betti number are

$$\begin{aligned} h_S^{4,0} &= 1, \\ h_S^{3,1} &= p_{\mathbf{w}}(d) - \sum_i p_{\mathbf{w}}(w_i), \\ h_S^{2,2} &= p_{\mathbf{w}}(2d) - \sum_i p_{\mathbf{w}}(d+w_i) + \sum_{i<j} p_{\mathbf{w}}(w_i+w_j). \end{aligned} \tag{6.1}$$

$$b_4(K_f) = 2 + 2h_S^{3,1} + h_S^{2,2}.$$

By Remark 3.14 these, together with $h_S^{1,3} = h_S^{3,1}$ and $h_S^{0,4} = 1$, are all the non-trivial Sasakian Hodge numbers: $h_S^{0,0} = 1$, every $h_S^{p,q}$ with $0 < p+q < 4$, such as $h_S^{1,1}$ and $h_S^{2,1}$, vanishes, and the harmonic forms of degree 5 to 9 are the Hodge duals of those of degree 4 to 0.

Computation. These numbers were computed for all 1 100 055 non-degenerate Calabi–Yau weight systems of [LSW1999], as listed by Kreuzer and Skarke [KS2026b], whose degrees range up to $d = 6\,521\,466$. Each Hodge number requires at most 22 denumerants $p_{\mathbf{w}}(k)$. These were computed exactly by enumerating the

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multiplicities of the four largest weights and counting the solutions for the two smallest weights in closed form. The number of $(x, y) \in \mathbb{Z}^2_{\geq 0}$ with $ax + by = r$ is zero unless $c = \gcd(a, b)$ divides r. Otherwise, put

$$a' = a/c, \qquad b' = b/c, \qquad r' = r/c,$$

and choose $0 \leq x_0 < b'$ satisfying

$$a'x_0 \equiv r' \pmod{b'}.$$

When $a'x_0 \leq r'$, the number of solutions is

$$\left\lfloor \frac{r' - a'x_0}{a'b'} \right\rfloor + 1. \tag{6.2}$$

The computation of all invariants, together with the twisted surfaces and the comparison (a) below, took 322 s of wall-clock time on a laptop with eight processes (Appendix A.1). By contrast, the Gröbner-basis approach of [AHHH+2024a] enumerates a monomial basis of the whole Milnor algebra, whose dimension $\mu(\mathbf{w})$ has median 1.1×10^7 over this dataset, exceeds 10^8 for 23% and 10^9 for 7% of the weight systems, and reaches 3.0×10^{12}. Gröbner-basis computations with the code of [AHHH+2024a] for these links had not finished after a month on a computing cluster, which these sizes explain.

Verification. Since no Gröbner-basis values were available in this dimension, the results were checked by four methods, which test different parts of the computation.

(a) *The Kreuzer–Skarke data.* The defect formula of Corollary 4.6, evaluated on the twisted surfaces, reproduces $h^{3,1}_{CY} - h^{3,1}_S$ exactly for all 1 100 055 systems, where $h^{3,1}_{CY}$ is the value computed in [LSW1999] with Vafa's formula and tabulated in [KS2026b]. The bound $h^{2,2}_S \leq h^{2,2}_{CY} - 1$ also holds in every case, with $h^{2,2}_{CY}$ obtained from (4.4). This compares $h^{3,1}_S$ with independently computed data, but it uses the same denumerant code as the computation.

(b) *Topology.* Milnor and Orlik computed the middle Betti number of a link from the monodromy of the Milnor fibration [MO1970]: writing $d/w_i = u_i/v_i$ in lowest terms,

$$b_{N-2}(K_f) = \sum_{I \subseteq \{1,\dots,N\}} (-1)^{N-|I|} \frac{\prod_{i \in I} u_i/v_i}{\operatorname{lcm}(u_i : i \in I)}.$$

This formula involves no denumerants and was implemented separately. It agrees with $b_4 = 2 + 2h^{3,1}_S + h^{2,2}_S$ for all 1 100 055 systems, and it reproduces $b_3 = 2 + 2h^{2,1}_S$ for all 7549 Gröbner-basis computations of [AHHH+2024a]. It checks the sum of the Hodge numbers rather than each of them; together with (a), it determines $h^{2,2}_S$ independently of its own formula.

(c) *Explicit polynomials.* These checks use neither the denumerant code nor Theorem 3.2. For each of the 42 weight systems of the authors' follow-up computations (file `CY4Polynomials_42.txt` of the GitHub repository), a Gröbner basis of $\mathcal{J}_f$ over $\mathbb{Q}$ was computed for an invertible polynomial (Lemma 3.16), and all standard monomials were enumerated. In every case their number is $\mu(\mathbf{w})$, the Hilbert function equals $P_{\mathbf{w}}(t)$ in every degree, and $(h^{4,0}_S, h^{3,1}_S, h^{2,2}_S)$ agrees with (6.1). For larger cases, $\dim(\mathbb{M}_f)_\ell$ was certified in single degrees by the exact certificate of Appendix A.1. In this way $h^{3,1}_S$ was certified for 900 weight systems that admit an invertible polynomial, 100 drawn at random from each range $10^k \leq \mu(\mathbf{w}) < 10^{k+1}$ with $4 \leq k \leq 12$, and $h^{2,2}_S$ for the 655 of them with $\dim S_{2d} \leq 4 \times 10^5$. Both numbers were also certified for 25 randomly drawn weight systems that admit no invertible polynomial and have $\dim S_{2d} \leq 2500$ and $\mu(\mathbf{w}) \leq 10^7$, using polynomials with all monomials of degree d and coefficients drawn uniformly from $\{1, \dots, 999\}$, and $h^{3,1}_S$ was certified for the 42 dense polynomials of the follow-up computations, which contain all monomials of degree d with coefficients between 1 and 4. The invertible polynomials are admissible by Lemma 3.16, and the dense ones were certified separately to have isolated singularities (Appendix A.1). Every certified value agrees with (6.1).

(d) *The sheaf-cohomology formula.* Formula (1.2) of Theorem B, evaluated term by term, reproduces both $h^{3,1}_S$ and $h^{2,2}_S$ for all 1 100 055 systems. This detects slips in the algebra of Corollary 3.6 and of the telescoping, but it shares the denumerant code with the computation.

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Results. Table 6.1 summarises the distribution of the invariants, and Table 6.2 lists a selection of links, including the extreme cases. Equality $h^{3,1}_S = h^{3,1}_{CY}$ holds for 534 935 weight systems (48.6%), which are exactly those with no twisted surface of positive geometric genus. Twisted surfaces occur in 930 088 systems. The defect is positive in the remaining 565 120 cases, with maximum 8500 for $\mathbb{P}(2,2,70,294,367,735)$, where the surface $X_f \cap \mathbb{P}(2,2,70,294)$, whose generic points have stabiliser μ_2, has $p_g = 8500$. Equality $h^{2,2}_S = h^{2,2}_{CY} - 1$ holds only for the sextic $\mathbb{P}(1,1,1,1,1,1)[6]$ and for $\mathbb{P}(1,1,1,1,1,5)[10]$; in both cases X_f is smooth, since it avoids the orbifold points of the ambient space, so that there are no twisted sectors. The pair $(h^{3,1}_S, h^{2,2}_S)$ takes 477 048 distinct values, plotted in Figure 6.1. The two numbers are strongly correlated: for the 364 766 weight systems with $h^{3,1}_S \geq 1000$ the ratio $h^{2,2}_S/h^{3,1}_S$ lies between 2.59 and 4.16, and it narrows to $[3.99, 4.01]$ for the 1273 systems with $h^{3,1}_S \geq 10^5$, so that for large $h^{3,1}_S$ the points approach the line $h^{2,2}_S = 4h^{3,1}_S$. This is explained by the identity (1.6) of Theorem E. The slope 4 is the coefficient of $h^{3,1}$ in the Libgober–Wood identity, a consequence of Riemann–Roch for $c_1 = 0$, and

$$\frac{h^{2,2}_S}{h^{3,1}_S} = 4 + \frac{47 + R}{h^{3,1}_S},$$

where R is a sum of Euler-characteristic-type invariants of the twisted sectors, which have dimension at most two, whereas $h^{3,1}_S$ is a Hodge number of the fourfold. Over the whole dataset $-10\,177 \leq R \leq 533$. The large negative values come from twisted surfaces, for which $6\chi(\mathcal{O}) - e$ is negative and of the order of their geometric genus. For $\mathbb{P}(2,2,90,281,750,1125)$, for instance, the surface $X_f \cap \mathbb{P}(2,2,90,750)$ has $p_g = 7236$ and contributes $-10\,485$ to $R = -10\,177$, so that $h^{2,2}_S/h^{3,1}_S = 140\,494/37\,656 \approx 3.73$. A twisted surface of positive genus lies in a weighted projective space $\mathbb{P}(\mathbf{w}_F)$ that omits two weights w_a, w_b not divisible by g_F, and its genus $p_g = p_{\mathbf{w}_F}(w_a + w_b)$ counts monomials of the small degree $w_a + w_b$. For the 1273 links with $h^{3,1}_S \geq 10^5$, whose smallest weight is 1 or 2, these genera are small, with $\sum_F (g_F - 1)p_g \leq 116$, and $-146 \leq 47 + R \leq 556$, which is less than 0.5% of $h^{3,1}_S$. Since there are only finitely many weight systems, this approach to the line is a property of the dataset rather than a limit theorem.

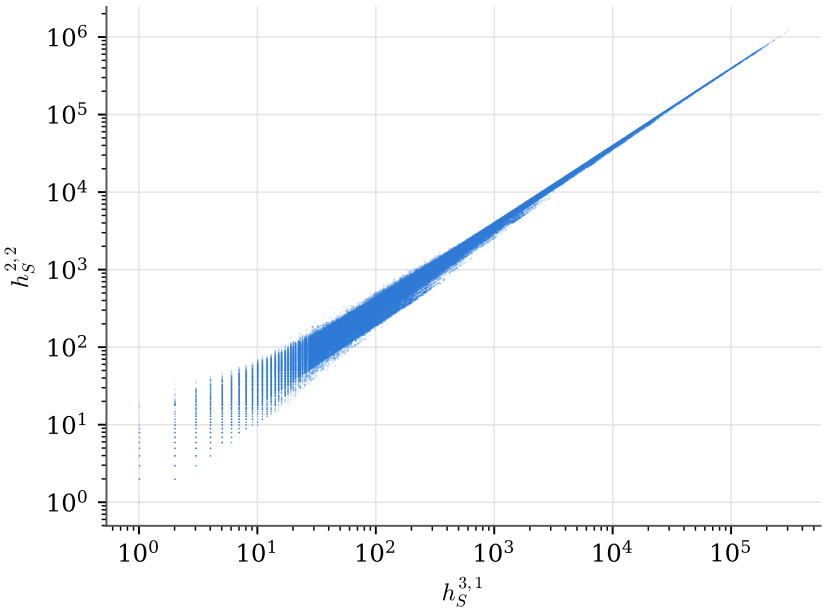


Figure 6.1: The Sasakian Hodge numbers $(h^{3,1}_S, h^{2,2}_S)$ of the 9-dimensional links of the 1 100 055 weighted Calabi–Yau fourfolds, on logarithmic axes; each point is a weight system, drawn semi-transparently so that darker regions contain more of them. The remaining Sasakian Hodge numbers are $h^{0,0}_S = h^{4,0}_S = 1$ and zero (Remark 3.14).

The full dataset, containing for each weight system its degree, the Kreuzer–Skarke Hodge numbers, $h^{3,1}_S$, $h^{2,2}_S$, $b_4(K_f)$, the exact value of $\mu(\mathbf{w})$ and the twisted surfaces, is available in the GitHub repository.

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Quantity	Minimum	Median	Maximum	Mean
degree d	6	4480	6 521 466	30 866
$h_S^{3,1}$	1	448	303 148	2268
$h_S^{2,2}$	1	1715	1 213 142	8989
$b_4(K_f)$	5	2610	1 819 440	13 526
$h_{CY}^{3,1} - h_S^{3,1}$	0	1	8500	32.6
$\mu(\mathbf{w})$	1.6×10^4	1.1×10^7	3.0×10^{12}	1.2×10^9

Table 6.1: Summary of the Sasakian Hodge numbers of the 1 100 055 nine-dimensional Calabi–Yau links, together with the degree, the defect of Corollary 4.6 and the Milnor number. Means are rounded to the nearest integer, except for the defect (one decimal) and $\mu(\mathbf{w})$, which like its other entries is given to two significant figures. The Hodge numbers are spread over several orders of magnitude, while the defect is usually small: its median is 1 and it vanishes for 48.6% of the links.

$\mathbf{w}$	d	$h_S^{3,1}$	$h_S^{2,2}$	$b_4(K_f)$	$h_{CY}^{3,1}$	$h_{CY}^{2,2}$	$\mu(\mathbf{w})$
$(1,1,1,1,1,1)$	6	426	1751	2605	426	1752	1.6×10^4
$(1,1,1,1,1,2)$	7	455	1866	2778	455	1868	1.9×10^4
$(1,1,1,1,2,2)$	8	443	1815	2703	443	1820	2.2×10^4
$(1,1,2,2,2,2)$	10	346	1406	2100	350	1452	2.1×10^4
$(2,2,2,3,3,3)$	15	187	788	1164	187	792	1.8×10^4
$(2,2,70,294,367,735)$	1470	20804	75830	117440	29304	117772	1.3×10^8
$(1,1,84,516,1204,1806)$	3612	303148	1213142	1819440	303148	1213644	6.6×10^9
$(105,230,279,462,534,805)$	2415	1	18	22	1	1140	4.8×10^4
$(521,600,624,625,630,750)$	3750	1	1	5	1	1752	1.6×10^4
$(1805,1806,151662,931638,2173822,3260733)$	6521466	252	502	1008	252	1213644	6.6×10^9
$(1,1806,75894,466206,1087814,1631721)$	3263442	151700	606822	910224	151700	1213644	3.0×10^{12}

Table 6.2: Selected nine-dimensional Calabi–Yau links, with the stringy Hodge numbers of the fourfold and the Milnor number. The rows comprise the sextic and four further links of small degree, among them $(1,1,2,2,2,2)$, whose twisted quintic surface $X_f \cap \mathbb{P}(2,2,2,2)$ has $p_g = 4$ and accounts for the defect of 4; the link with the largest defect; the maxima of $h_S^{3,1}$, $h_S^{2,2}$, d and μ; and links with $h_S^{3,1} = 1$ and with $h_S^{2,2} = 1$, the latter over the mirror of the sextic, with $(h_{CY}^{1,1}, h_{CY}^{3,1}) = (426, 1)$.

7 Conclusions

We conclude by relating the structural results to the computational observations that motivated them and by indicating the remaining topological questions. The two conjectures of [AHHH+2024a] are proved in Theorem A and Theorem D. The Weight Determinacy Theorem makes the isolated-singularity hypothesis explicit and holds in every degree of the Milnor algebra, because the Hilbert series of a weighted homogeneous isolated singularity is determined by its weights. Theorem B gives an alternative proof of the Milnor-algebra description used in [AHHH+2024a], through sheaf cohomology on the leaf space. Theorem C bounds the middle Sasakian Hodge numbers by the stringy Hodge numbers in every dimension $n \geq 2$, with a correction on the diagonal; for $n \geq 3$ the gap in $h^{n-1,1}$ is an explicit sum of geometric genera of twisted sectors, and for threefolds the bound is sharp exactly when all curves of orbifold points are rational.

The machine-learning results of [AHHH+2024a] can be read in this light. The Sasakian Hodge number $h_S^{2,1}$ is an alternating sum of six lattice-point counts in the weights, which may be part of the reason why it could be learnt accurately, although such counts depend on arithmetic properties of the weights and are not smooth functions of them. Its correlation with $h_{CY}^{2,1}$ is explained by the defect, which vanishes for 56% of the weight systems. The same formulae give complete datasets of Sasakian Hodge numbers, here for all 95 K3 links and all 1 100 055 Calabi–Yau fourfold links, the latter beyond the reach of the Gröbner-basis pipeline of [AHHH+2024a]. These datasets are complete in a strong sense, since all Sasakian Hodge numbers outside the middle degree are trivial. For the same reason they cannot exhibit mirror symmetry: the links see only the untwisted sector of the stringy Hodge numbers, and mirror symmetry exchanges the polynomial deformations counted by $h_S^{n-1,1}$ with classes of type $(1,1)$ on the mirror that come from its twisted sectors (Remark 4.7). Natural next steps are to determine the torsion and diffeomorphism types of the 7- and 9-dimensional links, the 5-dimensional ones being classified [BGM2006, S1962]; extend the Sasakian Hodge number database to higher dimensional links; to look for analogues of the Crowley–Nordström invariant for the SU(n)-structures

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of contact Calabi–Yau manifolds of higher dimension; to compare the invariants of the links, including ν, for pairs of weight systems whose Calabi–Yau hypersurfaces are mirror to each other; and to use the new datasets as testing grounds for machine learning in geometry.

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AI usage

Claude (Anthropic) assisted with the computations and the first draft of this article. Codex (OpenAI) assisted with revising the exposition, organisation, internal references, bibliography and LaTeX source, and with checking compilation and layout. The authors retain responsibility for the mathematical content, computational results and references.

A Certificates and computations

This appendix records the computational checks used in Sections 3.3, 3.4, 5 and 6, together with the computing environment and the complete K3 dataset. We retain the weighted grading $\deg z_i = w_i$ and the Jacobian ideal $\mathcal{J}_f$ of Section 2.1; admissibility of the invertible representatives is established in Lemma 3.16. The code, the input data with their sources and SHA-256 hashes, the complete output datasets, the test polynomials with their random seeds, and the logs of the computations described here are available in the GitHub repository, from which every table and count in this paper can be regenerated. The inputs are the weight systems and Kreuzer–Skarke Hodge numbers of the 7555 threefolds as used in [AHHH+2024a] (files `WP4s.txt` and `WP4_Hodges.txt` of [AHHH+2024b]), the 7549 Gröbner-basis computations of [AHHH+2024a], and the fourfold list `5dTransWH.all.gz` of [KS2026b], whose Hodge numbers are the chiral-ring data computed with Vafa's formula.

Checks of the sheaf-cohomological formula. To check the algebra in the proof of Theorem B, (1.2) was evaluated term by term, without using the telescoping. The results were compared with:

1. the Hodge numbers of nine smooth hypersurfaces in projective space, including the cubic fourfold with $(h^{3,1}, h^{2,2}_{\mathrm{prim}}) = (1, 20)$;
2. the coefficients of $P_{\mathbf{w}}$ for 3000 Hodge numbers of random non-degenerate weight systems with $N = 3, \dots, 6$, most of them not Calabi–Yau;
3. Gröbner-basis computations over $\mathbb{Q}$ of the Milnor algebras of invertible polynomials (Lemma 3.16) for 300 of these systems;
4. all Sasakian Hodge numbers in Sections 5 and 6.

For $N = 3$, only the formula is tested: Theorem B assumes $n \geq 2$ and therefore $N \geq 4$. No discrepancy was found.

A.1 Certificates for explicit polynomials

The checks below address two separate requirements: the critical point must be isolated, and the relevant graded piece must have the stated dimension. The first is automatic for the representatives of Lemma 3.16; dense polynomials require a separate isolation test.

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Isolation certificates. The remaining test polynomials contain all monomials of degree d: the $6+25$ polynomials with coefficients drawn uniformly from $\{1,\dots,999\}$ in Sections 3.4 and 6, and the $95+42$ polynomials of the follow-up computations, whose coefficients lie between 1 and 4 and are mostly equal to 1. For such a polynomial f a Gröbner basis of the Jacobian ideal of the reduction $\bar f$ of f modulo $p = 32003$ was computed with Singular [DGPS2025], and the quotient was checked to have Krull dimension 0 and vector-space dimension $\mu(\mathbf{w})$. This implies that f itself has an isolated critical point at the origin. Indeed, in every degree ℓ,

$$\dim_{\mathbb{Q}}(\mathbb{M}_f)_\ell \le \dim_{\mathbb{F}_p}(\mathbb{M}_{\bar f})_\ell, \tag{A.1}$$

because the rank of the integer matrix of the map

$$\bigoplus_i S_{\ell-d+w_i} \longrightarrow S_\ell, \qquad (g_i) \longmapsto \sum_i g_i \partial_i f,$$

can only drop modulo p. Thus (A.1) shows that $\mathbb{M}_f$ is finite-dimensional if $\mathbb{M}_{\bar f}$ is. Isolation was certified in this way for the six polynomials of Section 3.4, the 95 K3 polynomials of Section 5, the 25 random fourfold polynomials of Section 6(c), with $\mu(\mathbf{w})$ up to 9.2×10^6, and the 42 fourfold polynomials of the follow-up computations. No polynomial failed the test, and none was discarded or resampled.

Exact certificates for graded pieces. For an integer polynomial f and a degree ℓ, the degree-ℓ strand of the Koszul complex of the partial derivatives of f is

$$K_{2,\ell} \xrightarrow{D_2} K_{1,\ell} \xrightarrow{D_1} K_{0,\ell} = S_\ell, \tag{A.2}$$

with

$$K_{1,\ell} = \bigoplus_i S_{\ell-(d-w_i)}, \qquad K_{2,\ell} = \bigoplus_{i<j} S_{\ell-(2d-w_i-w_j)},$$

where $D_1(g_i)_i = \sum_i g_i\partial_i f$ and D_2 maps the generator e_{ij} to $\partial_i f\, e_j - \partial_j f\, e_i$. In the monomial bases these are integer matrices with $D_1D_2 = 0$, and $(\mathbb{M}_f)_\ell = S_\ell/\operatorname{im} D_1$. Let r_k be the rank of D_k modulo a prime p, and $c_1 = \dim K_{1,\ell}$. Since $r_k \le \operatorname{rank}_{\mathbb{Q}} D_k$ and $\operatorname{rank}_{\mathbb{Q}} D_1 + \operatorname{rank}_{\mathbb{Q}} D_2 \le c_1$, the condition

$$r_1 + r_2 = c_1 \tag{A.3}$$

forces $\operatorname{rank}_{\mathbb{Q}} D_k = r_k$ for both k, and then

$$\dim_{\mathbb{Q}}(\mathbb{M}_f)_\ell = \dim S_\ell - r_1$$

exactly. For an admissible f the Koszul complex is exact by Theorem 3.2, so (A.3) holds for all but finitely many primes. It was tested with exact Gaussian elimination modulo $p = 2^{31}-1$, without floating-point arithmetic, and it held at this prime in every case reported here, so that no other prime was needed. For $\ell = d$ the space $K_{2,\ell}$ is zero in the Calabi–Yau case, and the condition is that D_1 has full rank. The largest strands certified have $\dim S_\ell = 1.8\times10^5$ for $\ell = d$ and 3.8×10^5 for $\ell = 2d$; the certificate for $h^{2,2}_S$ was attempted when $\dim S_{2d} \le 4\times10^5$. The certificate does not by itself show that f is admissible, which is established separately by Lemma 3.16 or Appendix A.1.

Table A.1 lists the checks with explicit polynomials. The 42 fourfold weight systems of the follow-up computations appear twice, with an invertible and with a dense polynomial; the other samples are disjoint.

Computing environment. The symbolic and numerical verifications reported in the manuscript used Python (`sympy`, `numpy`), Singular [DGPS2025] and Macaulay2 [GS2026]. Unless stated otherwise, the computations were run with Python 3.10.13 and `numpy` 1.26.4, and the reruns recorded in the GitHub repository with `numpy` 2.2.6, on an Apple M1 laptop with eight cores and 16 GB of memory under macOS 26, using exact integer arithmetic throughout. The 322 s for the fourfold dataset is the wall-clock time of eight worker processes that compute all invariants of Section 6, the twisted surfaces and the comparison (a), including writing the output file and excluding reading the input file. The certificates of Section 3.4 took about 80 s on one core, and those of Section 6(c) took 69 s for the 900 systems with invertible polynomials and 15 s for the 25 systems without, with eight processes. The isolation certificates were computed with Singular 4.4.1 [DGPS2025] on the same machine while it was shared with other computations; individual polynomials took between a fraction of a second and 86 minutes.

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Section	n	Polynomials	Isolation	Quantity certified	Method
3.3	1–4	300 invertible, $\mu < 4000$	Lemma 3.16	all middle $h_S^{p,q}$	Gröbner basis over $\mathbb{Q}$
3.4	3	6 invertible	Lemma 3.16	$\dim(\mathbb{M}_f)_\ell$, $0 \le \ell \le 3d$	rank certificate
3.4	3	6 dense, random	Singular	$h_S^{2,1}$	rank certificate
5	2	95 dense	Singular	$h_S^{1,1}$	Gröbner basis
6	4	42 invertible	Lemma 3.16	Hilbert function	Gröbner basis over $\mathbb{Q}$
6	4	900 invertible	Lemma 3.16	$h_S^{3,1}$ (900), $h_S^{2,2}$ (655)	rank certificate
6	4	25 dense, random	Singular	$h_S^{3,1}$, $h_S^{2,2}$	rank certificate
6	4	42 dense	Singular	$h_S^{3,1}$	rank certificate

Table A.1: Checks with explicit polynomials. All certified values agree with the closed formulae.

A.2 Complete K3 dataset

Table A.2 contains the 95 K3 links discussed in Section 5. The Hodge and Betti numbers use (5.1), and the singularity column records the Du Val types entering Proposition 5.1. The table also recovers the topological classification recalled in Remark 5.2.

Table A.2: The 95 five-dimensional K3 links, ordered by degree and then lexicographically by weights: weights $\mathbf{w}$, degree d, Sasakian Hodge number $h_S^{1,1}$ from (5.1), second Betti number $b_2(K_f)$, singularities of the K3 surface X_f, and Milnor number $\mu(\mathbf{w})$. In every case $h_S^{2,0} = 1$, and the singularities have $19 - h_S^{1,1}$ exceptional curves in total (Proposition 5.1).

$\mathbf{w}$	d	$h_S^{1,1}$	b_2	Sing X_f	μ
$(1,1,1,1)$	4	19	21	–	81
$(1,1,1,2)$	5	18	20	A_1	96
$(1,1,1,3)$	6	19	21	–	125
$(1,1,2,2)$	6	16	18	$3A_1$	100
$(1,1,2,3)$	7	16	18	$A_2 + A_1$	120
$(1,1,2,4)$	8	17	19	$2A_1$	147
$(1,2,2,3)$	8	13	15	$A_2 + 4A_1$	105
$(1,1,3,4)$	9	16	18	A_3	160
$(1,2,3,3)$	9	12	14	$3A_2 + A_1$	112
$(1,1,3,5)$	10	17	19	A_2	189
$(1,2,2,5)$	10	14	16	$5A_1$	144
$(1,2,3,4)$	10	12	14	$A_3 + A_2 + 2A_1$	126
$(1,2,3,5)$	11	12	14	$A_4 + A_2 + A_1$	144
$(1,1,4,6)$	12	18	20	A_1	242
$(1,2,3,6)$	12	13	15	$2A_2 + 2A_1$	165
$(1,2,4,5)$	12	12	14	$A_4 + 3A_1$	154
$(1,3,4,4)$	12	10	12	$3A_3$	132
$(2,2,3,5)$	12	9	11	$A_4 + 6A_1$	105
$(2,3,3,4)$	12	8	10	$4A_2 + 3A_1$	90
$(1,3,4,5)$	13	10	12	$A_4 + A_3 + A_2$	144
$(1,2,4,7)$	14	13	15	$A_3 + 3A_1$	195
$(2,2,3,7)$	14	10	12	$A_2 + 7A_1$	132
$(2,3,4,5)$	14	7	9	$A_4 + A_3 + A_2 + 3A_1$	99
$(1,2,5,7)$	15	12	14	$A_6 + A_1$	208
$(1,3,4,7)$	15	10	12	$A_6 + A_3$	176
$(1,3,5,6)$	15	10	12	$A_5 + 2A_2$	168
$(2,3,5,5)$	15	6	8	$3A_4 + A_1$	104
$(3,3,4,5)$	15	6	8	$A_3 + 5A_2$	88
$(1,2,5,8)$	16	13	15	$A_4 + 2A_1$	231
$(1,3,4,8)$	16	11	13	$2A_3 + A_2$	195
$(1,4,5,6)$	16	9	11	$A_5 + A_4 + A_1$	165
$(2,3,4,7)$	16	7	9	$A_6 + A_2 + 4A_1$	117
$(2,3,5,7)$	17	6	8	$A_6 + A_4 + A_2 + A_1$	120
$(1,2,6,9)$	18	14	16	$A_2 + 3A_1$	272

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Table A.2 continued.

w	d	$h_S^{1,1}$	b_2	Sing X_f	μ
$(1,3,5,9)$	18	11	13	A_4+2A_2	221
$(1,4,6,7)$	18	9	11	$A_6+A_3+A_1$	187
$(2,3,4,9)$	18	8	10	$A_3+2A_2+4A_1$	140
$(2,3,5,8)$	18	6	8	$A_7+A_4+2A_1$	130
$(3,4,5,6)$	18	5	7	$A_4+A_3+3A_2+A_1$	91
$(3,4,5,7)$	19	4	6	$A_6+A_4+A_3+A_2$	96
$(1,4,5,10)$	20	10	12	$2A_4+A_1$	228
$(2,3,5,10)$	20	7	9	$2A_4+A_2+2A_1$	153
$(2,4,5,9)$	20	6	8	A_8+5A_1	132
$(2,5,6,7)$	20	5	7	$A_6+A_5+3A_1$	117
$(3,4,5,8)$	20	4	6	$A_7+2A_3+A_2$	102
$(1,3,7,10)$	21	10	12	A_9	264
$(1,5,7,8)$	21	8	10	A_7+A_4	208
$(2,3,7,9)$	21	6	8	$A_8+2A_2+A_1$	152
$(3,5,6,7)$	21	4	6	$A_5+A_4+3A_2$	96
$(1,3,7,11)$	22	11	13	A_6+A_2	285
$(1,4,6,11)$	22	10	12	$A_5+A_3+A_1$	252
$(2,4,5,11)$	22	7	9	$A_4+A_3+5A_1$	153
$(1,3,8,12)$	24	12	14	A_3+2A_2	322
$(1,6,8,9)$	24	8	10	$A_8+A_2+A_1$	230
$(2,3,7,12)$	24	7	9	$A_6+2A_2+2A_1$	187
$(2,3,8,11)$	24	6	8	$A_{10}+3A_1$	182
$(3,4,5,12)$	24	5	7	$A_4+2A_3+2A_2$	133
$(3,4,7,10)$	24	3	5	$A_9+A_6+A_1$	119
$(3,6,7,8)$	24	4	6	$A_6+4A_2+A_1$	102
$(4,5,6,9)$	24	3	5	$A_8+A_4+A_2+2A_1$	95
$(4,5,7,9)$	25	2	4	$A_8+A_6+A_3$	96
$(1,5,7,13)$	26	9	11	A_6+A_4	285
$(2,3,8,13)$	26	7	9	$A_7+A_2+3A_1$	207
$(2,5,6,13)$	26	6	8	$A_5+A_4+4A_1$	168
$(2,5,9,11)$	27	4	6	$A_{10}+A_4+A_1$	160
$(5,6,7,9)$	27	2	4	$A_6+A_5+A_4+A_2$	88
$(1,4,9,14)$	28	10	12	A_8+A_1	342
$(3,4,7,14)$	28	4	6	$2A_6+A_2+A_1$	150
$(4,6,7,11)$	28	2	4	$A_{10}+A_5+2A_1$	102
$(1,4,10,15)$	30	11	13	$A_4+A_3+A_1$	377
$(1,6,8,15)$	30	9	11	$A_7+A_2+A_1$	319
$(2,3,10,15)$	30	8	10	$A_4+2A_2+3A_1$	252
$(2,6,7,15)$	30	6	8	$A_6+A_2+5A_1$	184
$(3,4,10,13)$	30	3	5	$A_{12}+A_3+A_1$	153
$(4,5,6,15)$	30	4	6	$2A_4+A_3+A_2+2A_1$	130
$(5,6,8,11)$	30	1	3	$A_{10}+A_7+A_1$	95
$(2,5,9,16)$	32	5	7	$A_8+A_4+2A_1$	207
$(4,5,7,16)$	32	3	5	$A_6+A_4+2A_3$	135
$(3,5,11,14)$	33	2	4	$A_{13}+A_4$	152
$(3,4,10,17)$	34	4	6	$A_9+A_3+A_2+A_1$	186
$(4,6,7,17)$	34	3	5	$A_6+A_5+A_3+2A_1$	135
$(1,5,12,18)$	36	10	12	A_5+A_4	434
$(3,4,11,18)$	36	4	6	$A_{10}+2A_2+A_1$	200
$(7,8,9,12)$	36	1	3	$A_7+A_6+A_3+A_2$	87
$(3,5,11,19)$	38	3	5	$A_{10}+A_4+A_2$	189
$(5,6,8,19)$	38	2	4	$A_7+A_5+A_4+A_1$	132
$(5,7,8,20)$	40	2	4	$A_6+2A_4+A_3$	132
$(1,6,14,21)$	42	10	12	$A_6+A_2+A_1$	492
$(2,5,14,21)$	42	6	8	$A_6+A_4+3A_1$	296
$(3,4,14,21)$	42	5	7	$A_6+A_3+2A_2+A_1$	247

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Table A.2 continued.

$\mathbf{w}$	d	$h^{1,1}_S$	b_2	Sing X_f	μ
$(4,5,13,22)$	44	2	4	$A_{12}+A_4+A_1$	186
$(3,5,16,24)$	48	4	6	$A_7+A_4+2A_2$	258
$(7,8,10,25)$	50	1	3	$A_7+A_6+A_4+A_1$	129
$(4,5,18,27)$	54	3	5	$A_8+A_4+A_3+A_1$	245
$(5,6,22,33)$	66	2	4	$A_{10}+A_4+A_2+A_1$	244

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